
A Step by Step Mathematical Derivation and Tutorial on Kalman Filters

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Abstract

We present a step by step mathematical derivation of the Kalman filter using two different approaches. First, we consider the orthogonal projection method by means of vector-space optimization. Second, we derive the Kalman filter using Bayesian optimal filtering. We provide detailed proofs for both methods and each equation is expanded in detail.

1 Introduction

The Kalman filter, named after Rudolf E. Kalman, is still a highly useful algorithm today despite having been introduced more than 50 years ago. Its success can be attributed to it being an optimal estimator and its relatively straightforward and easy to implement recursive algorithm with small computational cost [3].

The Kalman filter has been used in various applications such as smoothing noisy data and providing estimates of parameters of interest, phase-locked loops in radio equipment, smoothing the output from laptop track pads, global positioning system receivers, and many others [10].

The Kalman filter [5], also known as the Kalman-Bucy filter [6], can be summarized as an iterative prediction-correction process. It can also be seen as a time variant Wiener filter [3] and was originally derived using the orthogonal projection method. The innovations approach [2] was developed in the late 1960s using martingales theory [9], [4].

In the first part of this article the orthogonal projection method is used to derive the Kalman filter as a minimum mean squared estimator. The

卡尔曼滤波器的逐步数学推导和教程

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Abstract

我们使用两种不同的方法逐步对卡尔曼滤波器进行数学推导。首先，我们考虑通过向量空间优化的正交投影方法。其次，我们使用贝叶斯最优过滤推导卡尔曼过滤器。我们为这两种方法提供了详细的证明，并且每个方程都得到了详细的扩展。

1 Introduction

卡尔曼滤波器以鲁道夫·E·卡尔曼 (Rudolf E. Kalman) 的名字命名，尽管已在 50 多年前推出，但至今仍然是一种非常有用的算法。它的成功可以归因于它是一个最优估计器，并且相对简单且易于实现计算成本较小的递归算法[3]。

卡尔曼滤波器已用于各种应用，例如平滑噪声数据并提供感兴趣的参数估计、无线电设备中的锁相环、平滑笔记本电脑触控板的输出、全球定位系统接收器以及许多其他应用[10]。

卡尔曼滤波器[5]，也称为卡尔曼-布西滤波器[6]，可以概括为迭代预测校正过程。它也可以被视为时变维纳滤波器 [3]，最初是使用正交投影方法推导出来的。创新方法 [2] 是在 20 世纪 60 年代末使用鞅理论 [9]、[4] 开发的。

在本文的第一部分中，使用正交投影方法来推导卡尔曼滤波器作为最小均方估计器。这

derivation is an expansion of the analysis presented in [1], such that each step of the proof is clearly derived and presented with complete details.

The Kalman filter has a Bayesian interpretation as well [7], [8] and can be derived within a Bayesian framework as a MAP estimator. The second part of this article uses Bayesian optimal filtering to derive the same equations.

推导是[1]中提出的分析的扩展，使得证明的每一步都清晰地推导并提供完整的细节。卡尔曼滤波器也具有贝叶斯解释[7]、[8]，并且可以在贝叶斯框架内作为 MAP 估计器导出。本文的第二部分使用贝叶斯最优过滤来推导相同的方程。

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2 Model of a Random Process

Consider that we have a target state vector $x_k \in \mathbb{R}^n$, where k is the time index. The target space evolves according to the discrete time stochastic model:

$$x_k = \phi_{k-1}(x_{k-1}, u_{k-1})$$

ϕ_{k-1} is a known, possibly nonlinear function of state x_{k-1} and u_{k-1} is the noise which counts e.g. for mis-modeling or disturbances in target motion.

Also consider that the measurements of the process (picked up by the sensor for example) are $z_k \in \mathbb{R}^m$. the measurements and states are related by

$$z_k = h_k(x_k, w_k)$$

where h_k is a known, possibly nonlinear function and w_k is the measurement noise.

w_k and u_{k-1} are assumed to be white with known probability distribution functions and independent of each other.

Filtering is an operation that involves extraction of information about a quantity of interest x_k at (discrete) time k by using data measured up to and including time k . Therefore, the objective of filtering is to recursively estimate x_k (target state) from the measurements z_k .

For the special case where ϕ_k and h_k are linear functions and the distribution of noise and initial states are Gaussian, the n -dimensional dynamic model of a random process reduces to the following linear/Gaussian model and consists of the following three parts:

1. A vector with difference equation

$$x_{k+1} = \Phi_k x_k + u_k \quad k = 0, 1, 2, \dots$$

which defines how the random vector x_k changes with time.

- Here x_k is an n -dimensional state vector where each component is a random variable.
- Φ_k is a known $n \times n$ matrix.
- u_k is an n -dim random vector of input with zero mean and there is zero correlation between present noise at the time k and past noise at time l , i.e:

$$E[u_k u_l'] = Q_k \delta_{kl} = \begin{cases} Q_k & k = l \\ 0 & k \neq l \end{cases}$$

where $Q_k > 0$ is a positive definite matrix.

2 Model of a Random Process

考虑我们有一个目标状态向量 $x_k \in \mathbb{R}^n$ ，其中 k 是时间索引。目标空间根据离散时间随机模型演化：

$$x_k = \phi_{k-1}(x_{k-1}, u_{k-1})$$

ϕ_{k-1} 是状态 x_{k-1} 的已知可能非线性函数，而 u_{k-1} 是计数的噪声，例如用于错误建模或目标运动的干扰。

还要考虑过程的测量值（例如由传感器拾取的）为 $z_k \in \mathbb{R}^m$ 。测量值和状态的关系为

$$z_k = h_k(x_k, w_k)$$

其中 h_k 是已知的可能非线性函数， w_k 是测量噪声。

w_k 和 u_{k-1} 假设为白色，具有已知的概率分布函数并且彼此独立。

过滤是涉及通过使用直到时间 k （包括时间 k ）测量的数据来提取关于在（离散）时间 k 的感兴趣量 x_k 的信息的操作。因此，过滤的目标是根据测量值 z_k 递归地估计 x_k （目标状态）。

对于 ϕ_k 和 h_k 为线性函数且噪声和初始状态分布为高斯分布的特殊情况，随机过程的 n 维动态模型可简化为以下线性/高斯模型，并由以下三部分组成：

1. 具有差分方程的向量

$$x_{k+1} = \Phi_k x_k + u_k \quad k = 0, 1, 2, \dots$$

它定义了随机向量 x_k 如何随时间变化。

- 这里 x_k 是一个 n 维状态向量，其中每个分量都是随机变量。
- Φ_k 是已知的 $n \times n$ 矩阵。
- u_k 是一个具有零均值的 n -dim 随机输入向量，并且 k 时刻的当前噪声与 l 时刻的过去噪声之间的相关性为零，即：

$$E[u_k u_l'] = Q_k \delta_{kl} = \begin{cases} Q_k & k = l \\ 0 & k \neq l \end{cases}$$

其中 $Q_k > 0$ 是正定矩阵。

2. An initial random vector x_0 and initial random estimate $\hat{x}_0$ with initial error covariance $E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)'] = P_0$
3. Measurements of the process is of the form

$$z_k = H_k x_k + w_k \quad k = 0, 1, 2, \dots$$

which defines how the measurements z_k of the process x_k are measured over time.

- Here H_k is a known $m \times n$ matrix
- w_k is an n -dimensional random measurement error with zero mean and

$$E[w_k w_l'] = R_k \delta_{kl} = \begin{cases} R_k & k = l \\ 0 & k \neq l \end{cases}$$

where $R_k > 0$ is a positive definite matrix.

It is assumed that x_0, u_j, w_k are all uncorrelated for $j \geq 0, k \geq 0$.

Part I

Derivation Using Vector Space Methods

3 Hilbert Space of Random Vectors

3.1 A Review of Probability

For a real valued random variable x , we define the probability distribution P of x by

$$P(\zeta) = \text{Prob}(x \leq \zeta).$$

In other words, $P(\zeta)$ is the probability that the random variable x assumes a value less than or equal to the number ζ . For a finite collection of real random variables $\{x_1, x_2, \dots, x_n\}$, their joint probability distribution P shows their inter-dependencies and is defined as

$$P(\zeta_1, \zeta_2, \dots, \zeta_n) = \text{Prob}(x_1 \leq \zeta_1, x_2 \leq \zeta_2, \dots, x_n \leq \zeta_n).$$

2. 初始随机向量 x_0 和具有初始误差协方差 $E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)'] = P_0$ 的初始随机估计 $\hat{x}_0$

3. 过程测量的形式为

$$z_k = H_k x_k + w_k \quad k = 0, 1, 2, \dots$$

它定义了如何随时间测量过程 x_k 的测量值 z_k 。

- 这里 H_k 是已知的 $m \times n$ 矩阵
- w_k 是均值为零的 n 维随机测量误差,

$$E[w_k w_l'] = R_k \delta_{kl} = \begin{cases} R_k & k = l \\ 0 & k \neq l \end{cases}$$

其中 $R_k > 0$ 是正定矩阵。

假设 x_0 、 u_j 、 w_k 对于 $j \geq 0, k \geq 0$ 都是不相关的。

Part I

Derivation Using Vector Space Methods

3 Hilbert Space of Random Vectors

3.1 A Review of Probability

对于实值随机变量 x ，我们将 x 的概率分布 P 定义为

$$P(\zeta) = \text{Prob}(x \leq \zeta).$$

换句话说， $P(\zeta)$ 是随机变量 x 取小于或等于数字 ζ 的值的概率。对于真实随机变量 $\{x_1, x_2, \dots, x_n\}$ 的有限集合，它们的联合概率分布 P 显示了它们的相互依赖关系，定义为

$$P(\zeta_1, \zeta_2, \dots, \zeta_n) = \text{Prob}(x_1 \leq \zeta_1, x_2 \leq \zeta_2, \dots, x_n \leq \zeta_n).$$

It is often useful to characterize a random variable by its mean and variance. Therefore the following quantities are of primary interest.

$$\begin{aligned} E[x] & \text{ is the } \textit{expected value} \text{ of } x. \\ E[x^2] & \text{ is the } \textit{average power} \text{ of } x. \\ E[(x - E(x))^2] & \text{ is the } \textit{variance} \text{ of } x. \end{aligned}$$

Note that the mathematical expectation operator $E[x]$ is a linear operator.

If $g(\cdot)$ is a single valued function then $g(x)$ is also a random variable and its expected value is defined as

$$E[g(x)] = \int_{-\infty}^{\infty} g(\zeta) dP(\zeta),$$

which may in general not be finite. Also, the expected value of any function $g(\cdot)$ over a collection of random variables $\{x_1, x_2, \dots, x_n\}$ is defined as

$$E[g(x_1, x_2, \dots, x_n)] = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} g(\zeta_1, \zeta_2, \dots, \zeta_n) dP(\zeta_1, \zeta_2, \dots, \zeta_n).$$

The second-order statistical averages of these variables can be described in terms of expected values. Specifically, for the $n \times n$ *covariance matrix* $cov(x_1, x_2, \dots, x_n)$, its ij -th element is defined as

$$E[(x_i - E(x_i))(x_j - E(x_j))] = E(x_i x_j) - E(x_i)E(x_j),$$

which in case of zero means reduces to $E(x_i x_j)$.

Finally, if $E(x_i x_j) = E(x_i)E(x_j)$ then $E[(x_i - E(x_i))(x_j - E(x_j))] = 0$ and we say that x_i and x_j are uncorrelated.

3.2 Random vectors

The idea of random variables can be generalized to random vectors. An n -dimensional random vector x is an ordered set of n random values x_i and is defined as

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Let $\{y_1, y_2, \dots, y_m\}$ be n -dimensional random vectors of the above form then a Hilbert space $\mathcal{H}$ can be defined such that $\mathcal{H}$ consists of all vectors

通过均值和方差来表征随机变量通常很有用。因此，以下数量是主要关注的。

$$\begin{aligned} E[x] & \text{ is the } \textit{expected value} \text{ of } x. \\ E[x^2] & \text{ is the } \textit{average power} \text{ of } x. \\ E[(x - E(x))^2] & \text{ is the } \textit{variance} \text{ of } x. \end{aligned}$$

请注意，数学期望运算符 $E[x]$ 是线性运算符。

如果 $g(\cdot)$ 是单值函数，则 $g(x)$ 也是随机变量，其期望值定义为

$$E[g(x)] = \int_{-\infty}^{\infty} g(\zeta) dP(\zeta),$$

一般来说，这可能不是有限的。此外，任意函数 $g(\cdot)$ 在随机变量集合 $\{x_1, x_2, \dots, x_n\}$ 上的期望值定义为

$$E[g(x_1, x_2, \dots, x_n)] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(\zeta_1, \zeta_2, \dots, \zeta_n) dP(\zeta_1, \zeta_2, \dots, \zeta_n).$$

这些变量的二阶统计平均值可以用期望值来描述。具体来说，对于 $n \times n$ covariance matrix $\text{cov}(x_1, x_2, \dots, x_n)$ ，其第 ij 个元素定义为

$$E[(x_i - E(x_i))(x_j - E(x_j))] = E(x_i x_j) - E(x_i)E(x_j),$$

在零的情况下意味着减少到 $E(x_i x_j)$ 。

最后，如果 $E(x_i x_j) = E(x_i)E(x_j)$ 则 $E[(x_i - E(x_i))(x_j - E(x_j))] = 0$ ，我们说 x_i 和 x_j 不相关。

3.2 Random vectors

随机变量的思想可以推广到随机向量。 n 维随机向量 x 是 n 随机值 x_i 的有序集合，定义为

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

令 $\{y_1, y_2, \dots, y_m\}$ 为上述形式的 n 维随机向量，则可以定义希尔伯特空间 $\mathcal{H}$ ，使得 $\mathcal{H}$ 由所有向量组成

whose components are linear combination of the y_i 's.

If x and y are elements of $\mathcal{H}$, we define their inner product as

$$(x|y) = E(x^T y) = E\left(\sum_{i=1}^n x_i y_i\right).$$

The induced norm of a vector x in this space can be written as

$$\|x\| = \sqrt{E(x^T x)} = \sqrt{E(x_1^2 + x_2^2 + \dots + x_n^2)} \quad (1)$$

$$= \sqrt{E(x_1^2) + E(x_2^2) + \dots + E(x_n^2)} \quad (2)$$

$$= \{\text{Trace}(E(xx^T))\}^{\frac{1}{2}}, \quad (3)$$

since the expected value of the random matrix xx^T is

$$E(xx^T) = \begin{bmatrix} E(x_1 x_1) & E(x_1 x_2) & \dots & E(x_1 x_n) \\ E(x_2 x_1) & E(x_2 x_2) & \dots & E(x_2 x_n) \\ \vdots & \vdots & \ddots & \vdots \\ E(x_n x_1) & E(x_n x_2) & \dots & E(x_n x_n) \end{bmatrix}.$$

Similarly the inner product can also be written as

$$(x|y) = \text{Trace}(E(xy^T)).$$

Two vectors are said to be orthogonal if $(x|y) = 0$ and this can be written as $x \perp y$. If x and y are uncorrelated and $E(x) = E(y) = 0$ then x and y are orthogonal to each other since

$$(x|y) = E(x^T y) = E(x)E(y) = 0 \Rightarrow x \perp y.$$

Finally, the covariance matrix for a random vector is defined as

$$\text{cov}(x) = E[(x - E(x))(x - E(x))^T].$$

If $E(x) = 0$ then the covariance matrix can be written as

$$E(xx^T).$$

其分量是 y_i 的线性组合。

如果 x 和 y 是 $\mathcal{H}$ 的元素，我们将它们的内积定义为

$$(x|y) = E(x^T y) = E\left(\sum_{i=1}^n x_i y_i\right).$$

该空间中向量 x 的归纳范数可以写为

$$\|x\| = \sqrt{E(x^T x)} = \sqrt{E(x_1^2 + x_2^2 + \dots + x_n^2)} \quad (1)$$

$$= \sqrt{E(x_1^2) + E(x_2^2) + \dots + E(x_n^2)} \quad (2)$$

$$= \{\text{Trace}(E(xx^T))\}^{\frac{1}{2}}, \quad (3)$$

因为随机矩阵 xx^T 的期望值为

$$E(xx^T) = \begin{bmatrix} E(x_1 x_1) & E(x_1 x_2) & \dots & E(x_1 x_n) \\ E(x_2 x_1) & E(x_2 x_2) & \dots & E(x_2 x_n) \\ \vdots & \vdots & \ddots & \vdots \\ E(x_n x_1) & E(x_n x_2) & \dots & E(x_n x_n) \end{bmatrix}.$$

类似地，内积也可以写成

$$(x|y) = \text{Trace}(E(xy^T)).$$

如果 $(x|y) = 0$ 则两个向量被称为正交，这可以写为 $x \perp y$ 。如果 x 和 y 不相关且 $E(x) = E(y) = 0$ ，则 x 和 y 彼此正交，因为

$$(x|y) = E(x^T y) = E(x)E(y) = 0 \Rightarrow x \perp y.$$

最后，随机向量的协方差矩阵定义为

$$\text{cov}(x) = E[(x - E(x))(x - E(x))^T].$$

如果 $E(x) = 0$ 则协方差矩阵可以写为

$$E(xx^T).$$

4 Minimum Variance Unbiased (Gauss-Markov) Estimate

4.1 Problem Setup

Assume that we have observations of a variable $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$ and that y is a linear estimation of other variables $W_{m \times n}$ plus an error term to account for measurement errors such that

$$y_i = \beta_1 w_{i1} + \beta_2 w_{i2} + \dots + \beta_n w_{in} + \varepsilon_i.$$

We can therefore write

$$y = W\beta + \varepsilon,$$

where y is the known outcome of m inexact measurements, W is a known matrix, β is an unknown vector of parameters and ε is a random vector such that

$$\begin{aligned} E(\varepsilon_i) &= 0, \\ E(\varepsilon\varepsilon') &= Q > 0, \end{aligned}$$

where Q is a positive definite matrix. Assuming W and y are known we want to estimate the unknown β . We seek a linear estimate $\hat{\beta} = Ky$, where $K_{m \times n}$ is an unknown constant.

Since ε is a random vector, $y = W\beta + \varepsilon$ is a random vector and since $\hat{\beta} = Ky$, $\hat{\beta}$ is also a random vector. As a result the estimation error defined by error = $\hat{\beta} - \beta$ is a random vector as well.

We consider the optimality criterion of minimizing the norm of the error in order to find $\hat{\beta}$. Since error is a random vector, the norm is defined as

$$\begin{aligned} \|\text{error}\|^2 &= E[(\hat{\beta} - \beta)^T(\hat{\beta} - \beta)] = E[(Ky - \beta)^T(Ky - \beta)] \\ &= E[(K(W\beta + \varepsilon) - \beta)^T(K(W\beta + \varepsilon) - \beta)] \\ &= E[(KW\beta + K\varepsilon - \beta)^T(KW\beta + K\varepsilon - \beta)] \\ &= E[((KW\beta)^T + (K\varepsilon)^T - \beta^T)(KW\beta + K\varepsilon - \beta)] \end{aligned}$$

after multiplying out

$$\begin{aligned} &= E[\underbrace{(KW\beta)^T(KW\beta)} + (KW\beta)^T(K\varepsilon) - \underbrace{(KW\beta)^T\beta} \\ &\quad + (K\varepsilon)^T(KW\beta) + (K\varepsilon)^T(K\varepsilon) - (K\varepsilon)^T\beta \\ &\quad - \underbrace{\beta^T(KW\beta)} - \beta^T(K\varepsilon) + \underbrace{\beta^T\beta}] \end{aligned}$$

4 Minimum Variance Unbiased (Gauss-Markov) Estimate

4.1 Problem Setup

假设我们观察到变量 $y = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{bmatrix}$ 并且 y 是

其他变量 $W_{m \times n}$ 的线性估计加上一个误差项来解释测量误差，使得

$$y_i = \beta_1 w_{i1} + \beta_2 w_{i2} + \dots + \beta_n w_{in} + \varepsilon_i.$$

因此我们可以写

$$y = W\beta + \varepsilon,$$

其中 y 是 m 不精确测量的已知结果， W 是已知矩阵， β 是未知参数向量， ε 是随机向量，使得

$$E(\varepsilon_i) = 0,$$

$$E(\varepsilon\varepsilon') = Q > 0,$$

其中 Q 是正定矩阵。假设 W 和 y 已知，我们想要估计未知的 β 。我们寻求线性估计 $\hat{\beta} = Ky$ ，其中 $K_{m \times n}$ 是未知常数。

由于 ε 是随机向量，因此 $y = W\beta + \varepsilon$ 是随机向量，并且由于 $\hat{\beta} = Ky$ ， $\hat{\beta}$ 也是随机向量。因此，由误差 $= \hat{\beta} - \beta$ 定义的估计误差也是一个随机向量。

为了找到 $\hat{\beta}$ ，我们考虑最小化误差范数的最优性标准。由于误差是一个随机向量，范数定义为

$$\begin{aligned} \|\text{error}\|^2 &= E[(\hat{\beta} - \beta)^T(\hat{\beta} - \beta)] = E[(Ky - \beta)^T(Ky - \beta)] \\ &= E[(K(W\beta + \varepsilon) - \beta)^T(K(W\beta + \varepsilon) - \beta)] \\ &= E[(KW\beta + K\varepsilon - \beta)^T(KW\beta + K\varepsilon - \beta)] \\ &= E[((KW\beta)^T + (K\varepsilon)^T - \beta^T)(KW\beta + K\varepsilon - \beta)] \end{aligned}$$

相乘后

$$\begin{aligned} &= E[\underbrace{(KW\beta)^T(KW\beta)} + \underbrace{(KW\beta)^T(K\varepsilon)} - \underbrace{(KW\beta)^T\beta} \\ &\quad + \underbrace{(K\varepsilon)^T(KW\beta)} + \underbrace{(K\varepsilon)^T(K\varepsilon)} - \underbrace{(K\varepsilon)^T\beta} \\ &\quad - \underbrace{\beta^T(KW\beta)} - \underbrace{\beta^T(K\varepsilon)} + \underbrace{\beta^T\beta}] \end{aligned}$$

separating the underlined terms

$$= E[(KW\beta)^T(KW\beta) + \beta^T\beta - (KW\beta)^T\beta - \beta^T(KW\beta)] \\ + E[\beta^T W^T K^T K \varepsilon + \varepsilon^T K^T KW \beta + (K\varepsilon)^T(K\varepsilon) - \varepsilon^T K^T \beta - \beta^T K \varepsilon]$$

and moving constant terms out of the expectation

$$= (KW\beta)^T(KW\beta) + \beta^T\beta - (KW\beta)^T\beta - \beta^T(KW\beta) \\ + \beta^T W^T K^T K E[\varepsilon] + E[\varepsilon^T] K^T KW \beta + E[(K\varepsilon)^T(K\varepsilon)] - E[\varepsilon^T] K^T \beta - \beta^T K E[\varepsilon].$$

Since the expected value of ε is zero then $E[\varepsilon] = E[\varepsilon^T] = 0$ and

$$\begin{aligned} \|\text{error}\|^2 &= \|KW\beta - \beta\|^2 + E[(K\varepsilon)^T(K\varepsilon)] \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(E[K\varepsilon(K\varepsilon)^T]) \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(E[K\varepsilon\varepsilon^T K^T]) \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(K E[\varepsilon\varepsilon^T] K^T) \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(K Q K^T). \end{aligned}$$

Note that we are trying to find the unknown K such that it minimizes the error. Yet, the error, in the expression above, is also a function of the unknown β . If $KW = I$ then the error expression is independent of β since

$$\|KW\beta - \beta\| = \|I\beta - \beta\| = 0.$$

The problem can now be written as

$$\begin{array}{ll} \arg \min_{\hat{\beta}} & \|\hat{\beta} - \beta\|^2 \\ \text{s.t. } KW = I & \end{array} = \begin{array}{ll} \arg \min_K & \text{Trace}(K Q K^T) \\ \text{s.t. } KW = I & \end{array}$$

which is independent of β .

4.2 What does imposing $KW = I$ mean?

We define the estimate $\hat{\beta}$ of an operator β to be unbiased if $E[\hat{\beta}] = \beta$. If we impose $KW = I$ then we can write

$$E[\hat{\beta}] = E[Ky] = E[KW\beta + K\varepsilon] = E[KW\beta] + E[K\varepsilon] = \underbrace{KW}_I \beta = \beta.$$

Therefore imposing $KW = I$ is equivalent to requiring that $\hat{\beta}$ be an unbiased estimate of β . In summary, we are trying to find the unbiased linear estimate of β that minimizes $\|\hat{\beta} - \beta\|^2$.

分隔下划线的术语

$$= E[(KW\beta)^T(KW\beta) + \beta^T\beta - (KW\beta)^T\beta - \beta^T(KW\beta)] \\ + E[\beta^TW^TK^TK\varepsilon + \varepsilon^TK^TKW\beta + (K\varepsilon)^T(K\varepsilon) - \varepsilon^TK^T\beta - \beta^TK\varepsilon]$$

并将常数项移出预期

$$= (KW\beta)^T(KW\beta) + \beta^T\beta - (KW\beta)^T\beta - \beta^T(KW\beta) \\ + \beta^TW^TK^TK E[\varepsilon] + E[\varepsilon^T]K^TKW\beta + E[(K\varepsilon)^T(K\varepsilon)] - E[\varepsilon^T]K^T\beta - \beta^TK E[\varepsilon].$$

由于 ε 的预期值为零, 则 $E[\varepsilon] = E[\varepsilon^T] = 0$ 且

$$\begin{aligned} \|\text{error}\|^2 &= \|KW\beta - \beta\|^2 + E[(K\varepsilon)^T(K\varepsilon)] \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(E[K\varepsilon(K\varepsilon)^T]) \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(E[K\varepsilon\varepsilon^TK^T]) \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(KE[\varepsilon\varepsilon^T]K^T) \\ &= \|KW\beta - \beta\|^2 + \text{Trace}(KQK^T). \end{aligned}$$

请注意, 我们试图找到未知的 K , 以便最大限度地减少误差。然而, 上面表达式中的误差也是未知的 β 的函数。如果 $KW = I$ 则错误表达式独立于 β , 因为

$$\|KW\beta - \beta\| = \|I\beta - \beta\| = 0.$$

现在问题可以写成

$$\begin{aligned} \arg \min_{\hat{\beta}} \quad & \|\hat{\beta} - \beta\|^2 \\ \text{s.t. } & KW = I \end{aligned} = \arg \min_K \quad \text{Trace}(KQK^T) \\ \text{s.t. } & KW = I$$

它独立于 β 。

4.2 What does imposing $KW = I$ mean?

如果 $E[\hat{\beta}] = \beta$, 我们将算子 β 的估计值 $\hat{\beta}$ 定义为无偏。如果我们施加 $KW = I$ 那么我们可以写

$$E[\hat{\beta}] = E[Ky] = E[KW\beta + K\varepsilon] = E[KW\beta] + E[K\varepsilon] = \underbrace{KW}_I\beta = \beta.$$

因此, 施加 $KW = I$ 相当于要求 $\hat{\beta}$ 是 β 的无偏估计。总之, 我们试图找到最小化 $\|\hat{\beta} - \beta\|^2$ 的 β 的无偏线性估计。

4.3 Solution to the problem

The minimization problem above can be written in terms of the elements of $\hat{\beta}$ as

$$\begin{aligned} \arg \min_{\hat{\beta}} \quad & \sum_{i=1}^n E[(\hat{\beta}_i - \beta_i)^2] \\ \text{s.t.} \quad & E[\hat{\beta}_i] = \beta_i, i = 1, 2, \dots, n \\ & \hat{\beta}_i = k'_i y, i = 1, 2, \dots, n \end{aligned}$$

Where k'_i is the i th row of the matrix K .

Since $E[(\hat{\beta}_i - \beta_i)^2] > 0$, every term in the above summation is nonnegative, therefore the sum $\sum_{i=1}^n E[(\hat{\beta}_i - \beta_i)^2]$ is minimum when each term $E[(\hat{\beta}_i - \beta_i)^2]$ is minimized. So we can solve n separate problems, one for each β_i as

$$\begin{aligned} \arg \min_{\hat{\beta}_i} \quad & E[(\hat{\beta}_i - \beta_i)^2] \\ \text{s.t.} \quad & E[\hat{\beta}_i] = \beta_i, i = 1, 2, \dots, n \\ & \hat{\beta}_i = k_i^T y, i = 1, 2, \dots, n \end{aligned}$$

We can also write the problem as finding the optimal matrix K

$$\begin{aligned} \arg \min_K \quad & \text{Trace}(KQK^T) \\ \text{s.t.} \quad & KW = I \end{aligned}$$

This can be thought of as a minimum weighted norm problem in the space of matrices, or it can also be decomposed into n separate problems where the i th problem is

$$\begin{aligned} \arg \min_{k_i} \quad & k_i^T Q k_i \\ \text{s.t.} \quad & k_i^T w_j = \delta_{ij}, i, j = 1, 2, \dots, n \end{aligned}$$

where w_j is the j th column of W , k_i is the i th row of K and δ_{ij} is the Kronecker delta function defined as

$$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Defining the weighted inner product as $(x|y)_Q = x^T Q y$ and noting that $(k_i|Q^{-1}w_j)_Q = k_i^T Q Q^{-1}w_j = k_i^T w_j$ the above problem can be written as

$$\begin{aligned} \arg \min_{k_i} \quad & (k_i|k_i)_Q \\ \text{s.t.} \quad & (k_i|Q^{-1}w_j)_Q = \delta_{ij} \end{aligned}$$

This is in the form of the standard minimum norm problem and can be

4.3 Solution to the problem

上面的最小化问题可以用 $\hat{\beta}$ 的元素写成

$$\begin{aligned} \arg \min_{\hat{\beta}} \quad & \sum_{i=1}^n E[(\hat{\beta}_i - \beta_i)^2] \\ \text{s.t.} \quad & E[\hat{\beta}_i] = \beta_i, i = 1, 2, \dots, n \\ & \hat{\beta}_i = k_i' y, i = 1, 2, \dots, n \end{aligned}$$

其中 k_i' 是矩阵 K 的第 i 行。

由于 $E[(\hat{\beta}_i - \beta_i)^2] > 0$, 上述求和中的每一项都是非负的, 因此当每一项 $E[(\hat{\beta}_i - \beta_i)^2]$ 最小化时, 总和 $\sum_{i=1}^n E[(\hat{\beta}_i - \beta_i)^2]$ 最小。因此我们可以解决 n 个单独的问题, 每个 β_i 一个问题如下

$$\begin{aligned} \arg \min_{\hat{\beta}_i} \quad & E[(\hat{\beta}_i - \beta_i)^2] \\ \text{s.t.} \quad & E[\hat{\beta}_i] = \beta_i, i = 1, 2, \dots, n \\ & \hat{\beta}_i = k_i^T y, i = 1, 2, \dots, n \end{aligned}$$

我们也可以将问题写成寻找最优矩阵 K

$$\begin{aligned} \arg \min_K \quad & \text{Trace}(KQK^T) \\ \text{s.t.} \quad & KW = I \end{aligned}$$

这可以被认为是矩阵空间中的最小加权范数问题, 或者它也可以分解为 n 个单独的问题, 其中第 i 个问题是

$$\begin{aligned} \arg \min_{k_i} \quad & k_i^T Q k_i \\ \text{s.t.} \quad & k_i^T w_j = \delta_{ij}, i, j = 1, 2, \dots, n \end{aligned}$$

其中 w_j 是 W 的第 j 列, k_i 是 K 的第 i 行, δ_{ij} 是克罗内克增量函数, 定义为

$$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

将加权内积定义为 $(x|y)_Q = x^T Q y$ 并注意到 $(k_i|Q^{-1}w_j)_Q = k_i^T Q Q^{-1}w_j = k_i^T w_j$ 上述问题可以写为

$$\begin{aligned} \arg \min_{k_i} \quad & (k_i|k_i)_Q \\ \text{s.t.} \quad & (k_i|Q^{-1}w_j)_Q = \delta_{ij} \end{aligned}$$

这是标准最小范数问题的形式, 可以是

rewritten as

$$\begin{aligned} \arg \min_{k_i} \quad & \|k_i\|_Q^2 \\ & k_i^T w_1 = 0 \\ & \vdots \\ \text{s.t.} \quad & k_i^T w_i = 1 \\ & \vdots \\ & k_i^T w_n = 0 \end{aligned} \quad \equiv \quad \arg \min_{k_i} \quad \|k_i\|_Q^2 \quad \text{s.t.} \quad \begin{bmatrix} -w_1^T \\ \vdots \\ -w_i^T \\ \vdots \\ -w_n^T \end{bmatrix} \begin{bmatrix} | \\ k_i \\ | \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} = e_i$$

The above problem can be summarized as

$$\begin{aligned} \arg \min_{k_i} \quad & \|k_i\|_Q^2 \\ \text{s.t.} \quad & W^T k_i = e_i \end{aligned}$$

Assuming that W is full column rank then W^T is full row rank and the least squares solution is

$$k_i = Q^{-1}W(W^TQ^{-1}W)^{-1}e_i.$$

We can now find K^T by combining all the k_i 's as

$$K^T = Q^{-1}W(W^TQ^{-1}W)^{-1}$$

and write the final solution $\hat{\beta}$ as

$$\hat{\beta} = Ky = (W^TQ^{-1}W)^{-1}W^TQ^{-1}y.$$

Here, we also compute the error covariance matrix as

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[(Ky - \beta)(Ky - \beta)^T] = E[(Ky - \beta)(y^TK^T - \beta^T)] \\ &= E[(KW\beta + K\varepsilon - \beta)(\beta^TW^TK^T + \varepsilon^TK^T - \beta^T)] \\ &= E[KW\beta\beta^TW^TK^T + KW\beta\varepsilon^TK^T - KW\beta\beta^T + K\varepsilon\beta^TW^TK^T \\ &\quad + K\varepsilon\varepsilon^TK^T - K\varepsilon\beta^T - \beta\beta^TW^TK^T - \beta\varepsilon^TK^T + \beta\beta^T] \\ &\stackrel{E[\varepsilon]=0}{=} E[(KW\beta - \beta)(KW\beta - \beta)^T] + K \underbrace{E[\varepsilon\varepsilon^T]}_Q K^T \end{aligned}$$

inserting $K = (W^TQ^{-1}W)^{-1}W^TQ^{-1}$

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[\underbrace{((W^TQ^{-1}W)^{-1}W^TQ^{-1}W)^{-1}}_{A^{-1}} \underbrace{W^TQ^{-1}W}_{A} \beta - \beta) (\underbrace{(W^TQ^{-1}W)^{-1}W^TQ^{-1}W}_{A^{-1}} \beta - \beta)^T] + KQK^T \\ &= \underbrace{E[(\beta - \beta)(\beta - \beta)^T]}_0 + \underbrace{(W^TQ^{-1}W)^{-1}W^TQ^{-1} \underbrace{QQ^{-1}}_I W}_{A} \underbrace{(W^TQ^{-1}W)^{-1}}_{A^{-1}} \\ &= (W^TQ^{-1}W)^{-1} \end{aligned}$$

重写为

$$\begin{aligned} \arg \min_{k_i} \quad & \|k_i\|_Q^2 \\ & k_i^T w_1 = 0 \\ & \vdots \\ \text{s.t.} \quad & k_i^T w_i = 1 \\ & \vdots \\ & k_i^T w_n = 0 \end{aligned} \quad \equiv \quad \arg \min_{k_i} \quad \|k_i\|_Q^2 \quad \text{s.t.} \quad \begin{bmatrix} -w_1^T \\ \vdots \\ -w_i^T \\ \vdots \\ -w_n^T \end{bmatrix} \begin{bmatrix} | \\ k_i \\ | \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} = e_i$$

上述问题可以概括为

$$\begin{aligned} \arg \min_{k_i} \quad & \|k_i\|_Q^2 \\ \text{s.t.} \quad & W^T k_i = e_i \end{aligned}$$

假设 W 是满列秩，则 W^T 是满行秩，最小二乘解为

$$k_i = Q^{-1}W(W^TQ^{-1}W)^{-1}e_i.$$

我们现在可以通过将所有 k_i 组合起来找到 K^T

$$K^T = Q^{-1}W(W^TQ^{-1}W)^{-1}$$

并将最终解 $\hat{\beta}$ 写为

$$\hat{\beta} = Ky = (W^TQ^{-1}W)^{-1}W^TQ^{-1}y.$$

在这里，我们还计算误差协方差矩阵：

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[(Ky - \beta)(Ky - \beta)^T] = E[(Ky - \beta)(y^TK^T - \beta^T)] \\ &= E[(KW\beta + K\varepsilon - \beta)(\beta^TW^TK^T + \varepsilon^TK^T - \beta^T)] \\ &= E[KW\beta\beta^TW^TK^T + KW\beta\varepsilon^TK^T - KW\beta\beta^T + K\varepsilon\beta^TW^TK^T \\ &\quad + K\varepsilon\varepsilon^TK^T - K\varepsilon\beta^T - \beta\beta^TW^TK^T - \beta\varepsilon^TK^T + \beta\beta^T] \\ &\stackrel{E[\varepsilon]=0}{=} E[(KW\beta - \beta)(KW\beta - \beta)^T] + K \underbrace{E[\varepsilon\varepsilon^T]}_Q K^T \end{aligned}$$

插入 $K = (W^TQ^{-1}W)^{-1}W^TQ^{-1}$

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[\underbrace{((W^TQ^{-1}W)^{-1}W^TQ^{-1}W)^{A^{-1}}}_A \underbrace{W^TQ^{-1}W}_A \beta - \beta) (\underbrace{(W^TQ^{-1}W)^{-1}W^TQ^{-1}W}_{A^{-1}} \beta - \beta)^T] + KQK^T \\ &= \underbrace{E[(\beta - \beta)(\beta - \beta)^T]}_0 + \underbrace{(W^TQ^{-1}W)^{-1}W^TQ^{-1} \underbrace{QQ^{-1}}_I W}_{A} \underbrace{(W^TQ^{-1}W)^{-1}}_{A^{-1}} \\ &= (W^TQ^{-1}W)^{-1} \end{aligned}$$

5 Minimum Variance Estimate

In the previous discussion β was assumed to be unknown and could take any value from $-\infty$ to $+\infty$. We had no prior knowledge about its values. If we have prior knowledge, such as β 's mean or covariance, then this prior info can be used to produce an estimate with lower error variance compared to the minimum variance unbiased estimate.

So, we assume that $y = W\beta + \varepsilon$ but in this case both ε and β are random vectors. We again want to find $\hat{\beta}$ such that we minimize the norm of the error.

5.1 Minimum Variance Estimate Theorem

Theorem 5.1. *Let y and β be random vectors. Assume that $[E[yy^T]]^{-1}$ exists. The linear estimate $\hat{\beta}$ of β based on y that minimizes $\|\hat{\beta} - \beta\|^2$ is*

$$\hat{\beta} = E[\beta y^T][E[yy^T]]^{-1}y,$$

with corresponding error covariance matrix

$$E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] = E[\beta\beta^T] - E[\hat{\beta}\hat{\beta}^T] = E[\beta\beta^T] - E[\beta y^T][E[yy^T]]^{-1}E[y\hat{\beta}].$$

Proof. Similar to the previous problem, this problem decomposes into a separate problem for each β_i . There are no constraints so we find the best approximation of β_i within the subspace generated by the y_i s.

Writing the optimal estimate as $\hat{\beta} = Ky$ where $K_{m \times n}$, then the i th subproblem is equivalent to the problem of selecting the i th row of K , which in turn gives the optimal linear combination of y_i s that make β_i s. So each row of K should satisfy the normal equations corresponding to projecting β_i onto the y_i . Specifically,

$$Ky = \begin{bmatrix} -k_1^T - \\ \vdots \\ -k_i^T - \\ \vdots \\ -k_n^T - \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_i \\ \vdots \\ \beta_n \end{bmatrix} = \beta$$

and $\text{error}_i = \beta_i - \hat{\beta}_i = \beta_i - k_i^T y$ should be orthogonal to each y_j (orthogonality principle), consequently

$$(\beta_i - k_i^T y | y_j) = 0 \Rightarrow (\beta_i | y_j) - (k_i^T y | y_j) = 0 \Rightarrow (k_i^T y | y_j) = (\beta_i | y_j)$$

5 Minimum Variance Estimate

在前面的讨论中， β 被假定为未知，并且可以采用从 $-\infty$ 到 $+\infty$ 的任何值。我们事先对其价值一无所知。如果我们有先验知识，例如 β 的均值或协方差，那么该先验信息可用于生成与最小方差无偏估计相比误差方差更低的估计。

因此，我们假设 $y = W\beta + \varepsilon$ ，但在本例中 ε 和 β 都是随机向量。我们再次想要找到 $\hat{\beta}$ ，以便最小化误差范数。

5.1 Minimum Variance Estimate Theorem

Theorem 5.1. *Let y and β be random vectors. Assume that $[E[yy^T]]^{-1}$ exists. The linear estimate $\hat{\beta}$ of β based on y that minimizes $\|\hat{\beta} - \beta\|^2$ is*

$$\hat{\beta} = E[\beta y^T][E[yy^T]]^{-1}y,$$

with corresponding error covariance matrix

$$E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] = E[\beta\beta^T] - E[\hat{\beta}\hat{\beta}^T] = E[\beta\beta^T] - E[\beta y^T][E[yy^T]]^{-1}E[y\hat{\beta}].$$

Proof. 与前一个问题类似，这个问题分解为每个 β_i 的单独问题。没有任何约束，因此我们在 y_i 生成的子空间内找到 β_i 的最佳近似。

将最优估计写为 $\hat{\beta} = Ky$ where $K_{m \times n}$ ，那么第 i 个子问题等价于选择 K 的第 i 行的问题，这又给出了构成 β_i 的 y_i 的最优线性组合。因此， K 的每一行都应满足将 β_i 投影到 y_i 上对应的正规方程。具体来说，

$$Ky = \begin{bmatrix} -k_1^T - \\ \vdots \\ -k_i^T - \\ \vdots \\ -k_n^T - \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_i \\ \vdots \\ \beta_n \end{bmatrix} = \beta$$

并且误差 $e_i = \beta_i - \hat{\beta}_i = \beta_i - k_i^T y$ 应该与每个 y_j (正交性原理)正交，因此

$$(\beta_i - k_i^T y | y_j) = 0 \Rightarrow (\beta_i | y_j) - (k_i^T y | y_j) = 0 \Rightarrow (k_i^T y | y_j) = (\beta_i | y_j)$$

$$\begin{aligned}
& \Rightarrow \begin{cases} (k_i^T y|y_1) = (\beta_i|y_1) \\ (k_i^T y|y_2) = (\beta_i|y_2) \\ \vdots \\ (k_i^T y|y_n) = (\beta_i|y_n) \end{cases} \Rightarrow \begin{cases} (k_{i1}y_1 + k_{i2}y_2 + \dots + k_{in}y_n|y_1) = (\beta_i|y_1) \\ (k_{i1}y_2 + k_{i2}y_2 + \dots + k_{in}y_n|y_2) = (\beta_i|y_2) \\ \vdots \\ (k_{i1}y_1 + k_{i2}y_2 + \dots + k_{in}y_n|y_n) = (\beta_i|y_n) \end{cases} \\
& \Rightarrow \begin{cases} k_{i1}(y_1|y_1) + k_{i2}(y_2|y_1) + \dots + k_{in}(y_n|y_1) = (\beta_i|y_1) \\ k_{i1}(y_1|y_2) + k_{i2}(y_2|y_2) + \dots + k_{in}(y_n|y_2) = (\beta_i|y_2) \\ \vdots \\ k_{i1}(y_1|y_n) + k_{i2}(y_2|y_n) + \dots + k_{in}(y_n|y_n) = (\beta_i|y_n) \end{cases} \\
& \Rightarrow \begin{cases} k_{i1}E[y_1y_1] + k_{i2}E[y_2y_1] + \dots + k_{in}E[y_ny_1] = E[\beta_i|y_1] \\ k_{i1}E[y_1y_2] + k_{i2}E[y_2y_2] + \dots + k_{in}E[y_ny_2] = E[\beta_i|y_2] \\ \vdots \\ k_{i1}E[y_1y_n] + k_{i2}E[y_2y_n] + \dots + k_{in}E[y_ny_n] = E[\beta_i|y_n] \end{cases} \\
& \Rightarrow \underbrace{\begin{bmatrix} E[y_1y_1] & E[y_2y_1] & \dots & E[y_ny_1] \\ E[y_1y_2] & E[y_2y_2] & \dots & E[y_ny_2] \\ \vdots & \ddots & & \vdots \\ E[y_1y_n] & E[y_2y_n] & \dots & E[y_ny_n] \end{bmatrix}}_{\text{symmetric}} \begin{bmatrix} k_{i1} \\ \vdots \\ k_{in} \end{bmatrix} = \begin{bmatrix} E[\beta_i|y_1] \\ \vdots \\ E[\beta_i|y_n] \end{bmatrix} \\
& \Rightarrow \underbrace{\begin{bmatrix} E[y_1y_1] & \dots & E[y_1y_n] \\ E[y_2y_1] & \dots & E[y_2y_n] \\ \vdots & \ddots & \vdots \\ E[y_ny_1] & \dots & E[y_ny_n] \end{bmatrix}}_{E[yy^T]} \underbrace{\begin{bmatrix} k_{i1} \\ \vdots \\ k_{in} \end{bmatrix}}_{k_i^T} = \begin{bmatrix} E[\beta_i|y_1] \\ \vdots \\ E[\beta_i|y_n] \end{bmatrix}
\end{aligned}$$

we have these matrix equations for every i , which can all be combined and written as

$$\begin{aligned}
E[yy^T]K^T &= E[y\beta^T] \Rightarrow K^T = [E[yy^T]]^{-1}E[y\beta^T] \\
K &= E[\beta y^T][E[yy^T]]^{-1}.
\end{aligned}$$

The error covariance matrix can now be written as

$$\begin{aligned}
E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[(Ky - \beta)(Ky - \beta)^T] = E[(Ky - \beta)(y^T K^T - \beta^T)] \\
&= E[Kyy^T K^T - Ky\beta^T - \beta y^T K^T + \beta\beta^T] \\
&= KE[yy^T]K^T - KE[y\beta^T] - E[\beta y^T]K^T + E[\beta\beta^T].
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \begin{cases} (k_i^T y|y_1) = (\beta_i|y_1) \\ (k_i^T y|y_2) = (\beta_i|y_2) \\ \vdots \\ (k_i^T y|y_n) = (\beta_i|y_n) \end{cases} \Rightarrow \begin{cases} (k_{i1}y_1 + k_{i2}y_2 + \dots + k_{in}y_n|y_1) = (\beta_i|y_1) \\ (k_{i1}y_2 + k_{i2}y_2 + \dots + k_{in}y_n|y_2) = (\beta_i|y_2) \\ \vdots \\ (k_{i1}y_1 + k_{i2}y_2 + \dots + k_{in}y_n|y_n) = (\beta_i|y_n) \end{cases} \\
&\Rightarrow \begin{cases} k_{i1}(y_1|y_1) + k_{i2}(y_2|y_1) + \dots + k_{in}(y_n|y_1) = (\beta_i|y_1) \\ k_{i1}(y_1|y_2) + k_{i2}(y_2|y_2) + \dots + k_{in}(y_n|y_2) = (\beta_i|y_2) \\ \vdots \\ k_{i1}(y_1|y_n) + k_{i2}(y_2|y_n) + \dots + k_{in}(y_n|y_n) = (\beta_i|y_n) \end{cases} \\
&\Rightarrow \begin{cases} k_{i1}E[y_1y_1] + k_{i2}E[y_2y_1] + \dots + k_{in}E[y_ny_1] = E[\beta_i|y_1] \\ k_{i1}E[y_1y_2] + k_{i2}E[y_2y_2] + \dots + k_{in}E[y_ny_2] = E[\beta_i|y_2] \\ \vdots \\ k_{i1}E[y_1y_n] + k_{i2}E[y_2y_n] + \dots + k_{in}E[y_ny_n] = E[\beta_i|y_n] \end{cases} \\
&\Rightarrow \underbrace{\begin{bmatrix} E[y_1y_1] & E[y_2y_1] & \dots & E[y_ny_1] \\ E[y_1y_2] & E[y_2y_2] & \dots & E[y_ny_2] \\ \vdots & \ddots & & \vdots \\ E[y_1y_n] & E[y_2y_n] & \dots & E[y_ny_n] \end{bmatrix}}_{\text{symmetric}} \begin{bmatrix} k_{i1} \\ \vdots \\ k_{in} \end{bmatrix} = \begin{bmatrix} E[\beta_i|y_1] \\ \vdots \\ E[\beta_i|y_n] \end{bmatrix} \\
&\Rightarrow \underbrace{\begin{bmatrix} E[y_1y_1] & \dots & E[y_1y_n] \\ E[y_2y_1] & \dots & E[y_2y_n] \\ \vdots & \ddots & \vdots \\ E[y_ny_1] & \dots & E[y_ny_n] \end{bmatrix}}_{E[yy^T]} \underbrace{\begin{bmatrix} k_{i1} \\ \vdots \\ k_{in} \end{bmatrix}}_{k_i^T} = \begin{bmatrix} E[\beta_i|y_1] \\ \vdots \\ E[\beta_i|y_n] \end{bmatrix}
\end{aligned}$$

对于每个 i ，我们都有这些矩阵方程，它们都可以组合并写为

$$E[yy^T]K^T = E[y\beta^T] \Rightarrow K^T = [E[yy^T]]^{-1}E[y\beta^T]$$

$$K = E[\beta y^T][E[yy^T]]^{-1}.$$

误差协方差矩阵现在可以写为

$$\begin{aligned}
E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[(Ky - \beta)(Ky - \beta)^T] = E[(Ky - \beta)(y^T K^T - \beta^T)] \\
&= E[Kyy^T K^T - Ky\beta^T - \beta y^T K^T + \beta\beta^T] \\
&= KE[yy^T]K^T - KE[y\beta^T] - E[\beta y^T]K^T + E[\beta\beta^T].
\end{aligned}$$

Noting that

$$\begin{aligned} K &= E[\beta y^T][E[yy^T]]^{-1} \\ K^T &= [E[yy^T]]^{-1} E[y\beta^T] \end{aligned}$$

and substituting for K and K^T we find

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[\beta y^T][E[yy^T]]^{-1} \underbrace{E[yy^T][E[yy^T]]^{-1}}_I E[y\beta^T] \\ &\quad - E[\beta y^T][E[yy^T]]^{-1} E[y\beta^T] - E[\beta y^T][E[yy^T]]^{-1} E[y\beta^T] + E[\beta\beta^T] \\ &= E[\beta\beta^T] - E[\beta y^T][E[yy^T]]^{-1} E[y\beta^T] \end{aligned}$$

□

If β and y have zero mean then

$$E[y] = 0 = E[W\beta + \varepsilon] = WE[\beta] + E[\varepsilon] \stackrel{E[\beta]=0}{=} E[\varepsilon] = 0$$

and we can write

$$E[\hat{\beta}] = E[Ky] = E[KW\beta + K\varepsilon] = KWE[\beta] + KE[\varepsilon] = 0 = E[\beta].$$

Therefore, $\hat{\beta}$ is an unbiased estimate of β .

Also, note that $\|\hat{\beta} - \beta\|^2$ can be written as

$$\|\hat{\beta} - \beta\|^2 = E[(\hat{\beta} - \beta)^T(\hat{\beta} - \beta)] = E[\|(\hat{\beta} - \beta)\|_2^2]$$

where $\|\cdot\|_2$ is the standard two-norm and we denote $E[\|(\hat{\beta} - \beta)\|_2^2]$ as the error variance.

Corollary 5.1. *Suppose that $y = W\beta + \varepsilon$, where y is a known m -dimensional vector, β is an n -dimensional unknown random vector, ε is an unknown m -dimensional random vector and $W_{m \times n}$ is a known constant matrix and*

$$E[\varepsilon\varepsilon^T] = Q \geq 0 \quad (\text{noise covariance})$$

$$E[\beta\beta^T] = R \geq 0 \quad (\text{input covariance for } \beta)$$

$$E[\varepsilon\beta^T] = 0 \quad (\text{no correlation between input and noise})$$

we also assume that $WRW^T + Q$ is invertible.

Then the linear estimate $\hat{\beta}$ of β that minimizes the error variance $E[\|\hat{\beta} - \beta\|_2^2]$ is

$$\hat{\beta} = RW^T(WRW^T + Q)^{-1}y \quad (4)$$

with error covariance

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = R - RW^T(WRW^T + Q)^{-1}WR$$

注意到

$$\begin{aligned} K &= E[\beta y^T][E[yy^T]]^{-1} \\ K^T &= [E[yy^T]]^{-1} E[y\beta^T] \end{aligned}$$

并替换 K 和 K^T 我们发现

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E[\beta y^T][E[yy^T]]^{-1} \underbrace{E[yy^T][E[yy^T]]^{-1}}_I E[y\beta^T] \\ &\quad - E[\beta y^T][E[yy^T]]^{-1} E[y\beta^T] - E[\beta y^T][E[yy^T]]^{-1} E[y\beta^T] + E[\beta\beta^T] \\ &= E[\beta\beta^T] - E[\beta y^T][E[yy^T]]^{-1} E[y\beta^T] \end{aligned}$$

□

如果 β 和 y 的均值为零, 则

$$E[y] = 0 = E[W\beta + \varepsilon] = WE[\beta] + E[\varepsilon] \stackrel{E[\beta]=0}{=} E[\varepsilon] = 0$$

我们可以写

$$E[\hat{\beta}] = E[Ky] = E[KW\beta + K\varepsilon] = KWE[\beta] + KE[\varepsilon] = 0 = E[\beta].$$

因此, $\hat{\beta}$ 是 β 的无偏估计。

另请注意, $\|\hat{\beta} - \beta\|^2$ 可以写为

$$\|\hat{\beta} - \beta\|^2 = E[(\hat{\beta} - \beta)^T(\hat{\beta} - \beta)] = E[\|(\hat{\beta} - \beta)\|_2^2]$$

其中 $\|\cdot\|_2$ 是标准二范数, 我们将 $E[\|(\hat{\beta} - \beta)\|_2^2]$ 表示为误差方差。

Corollary 5.1. Suppose that $y = W\beta + \varepsilon$, where y is a known m -dimensional vector, β is an n -dimensional unknown random vector, ε is an unknown m -dimensional random vector and $W_{m \times n}$ is a known constant matrix and

$$E[\varepsilon\varepsilon^T] = Q \geq 0 \quad (\text{noise covariance})$$

$$E[\beta\beta^T] = R \geq 0 \quad (\text{input covariance for } \beta)$$

$$E[\varepsilon\beta^T] = 0 \quad (\text{no correlation between input and noise})$$

we also assume that $WRW^T + Q$ is invertible.

Then the linear estimate $\hat{\beta}$ of β that minimizes the error variance $E[\|\hat{\beta} - \beta\|_2^2]$ is

$$\hat{\beta} = RW^T(WRW^T + Q)^{-1}y \quad (4)$$

with error covariance

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = R - RW^T(WRW^T + Q)^{-1}WR$$

Proof.

$$\begin{aligned}
E[yy^T] &= E[(W\beta + \varepsilon)(W\beta + \varepsilon)^T] \\
&= E[W\beta\beta^T W^T + W\beta\varepsilon^T + \varepsilon\beta^T W^T + \varepsilon\varepsilon^T] \\
&= W \underbrace{E[\beta\beta^T]}_R W^T + W \underbrace{E[\beta\varepsilon^T]}_0 + \underbrace{E[\varepsilon\beta^T]}_0 W^T + \underbrace{E[\varepsilon\varepsilon^T]}_Q \\
&= WRW^T + Q
\end{aligned}$$

and

$$\begin{aligned}
E[\beta y^T] &= E[\beta(W\beta + \varepsilon)^T] = E[\beta(\beta^T W^T + \varepsilon^T)] \\
&= E[\beta\beta^T W^T + \beta\varepsilon^T] = \underbrace{E[\beta\beta^T]}_R W^T + E[\beta\varepsilon^T] = RW^T,
\end{aligned}$$

therefore

$$\hat{\beta} = E[\beta y^T][E[yy^T]]^{-1} = RW^T(WRW^T + Q)^{-1}y$$

and

$$\begin{aligned}
E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] &= E[\beta\beta^T] - E[\beta y^T][E[yy^T]]^{-1}E[y\beta^T] \\
&= R - RW^T(WRW^T + Q)^{-1}WR^T \\
&= R - RW^T(WRW^T + Q)^{-1}WR,
\end{aligned}$$

since $E[y\beta^T] = WR^T = WR$. □

Corollary 5.2. *The estimate given by corollary 5.1 can be expressed in the alternative form*

$$\hat{\beta} = (W^T Q^{-1} W + R^{-1})^{-1} W^T Q^{-1} y \quad (5)$$

with corresponding error covariance

$$E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] = (W^T Q^{-1} W + R^{-1})^{-1}.$$

Proof. We need to show that $RW^T(WRW^T + Q)^{-1} = (W^T Q^{-1} W + R^{-1})^{-1} W^T Q^{-1}$. We prove this by pre-multiplying both sides by $(WRW^T + Q)$ and post-multiplying both sides by $(W^T Q^{-1} W + R^{-1})$.

$$\begin{aligned}
&(W^T Q^{-1} W + R^{-1})[RW^T(WRW^T + Q)^{-1}](WRW^T + Q) = \\
&(W^T Q^{-1} W + R^{-1})[(W^T Q^{-1} W + R^{-1})^{-1} W^T Q^{-1}](WRW^T + Q) \\
&\Leftrightarrow (W^T Q^{-1} W + R^{-1})RW^T = W^T Q^{-1}(WRW^T + Q) \\
&\Leftrightarrow W^T Q^{-1} WRW^T + \underbrace{R^{-1} R}_I W^T = W^T Q^{-1} WRW^T + W^T \underbrace{Q^{-1} Q}_I \\
&\Leftrightarrow W^T Q^{-1} WRW + W^T = W^T Q^{-1} WRW^T + W^T \checkmark
\end{aligned}$$

Proof.

$$\begin{aligned}
E[yy^T] &= E[(W\beta + \varepsilon)(W\beta + \varepsilon)^T] \\
&= E[W\beta\beta^T W^T + W\beta\varepsilon^T + \varepsilon\beta^T W^T + \varepsilon\varepsilon^T] \\
&= W \underbrace{E[\beta\beta^T]}_R W^T + W \underbrace{E[\beta\varepsilon^T]}_0 + \underbrace{E[\varepsilon\beta^T]}_0 W^T + \underbrace{E[\varepsilon\varepsilon^T]}_Q \\
&= WRW^T + Q
\end{aligned}$$

和

$$\begin{aligned}
E[\beta y^T] &= E[\beta(W\beta + \varepsilon)^T] = E[\beta(\beta^T W^T + \varepsilon^T)] \\
&= E[\beta\beta^T W^T + \beta\varepsilon^T] = \underbrace{E[\beta\beta^T]}_R W^T + E[\beta\varepsilon^T] = RW^T,
\end{aligned}$$

所以

$$\hat{\beta} = E[\beta y^T][E[yy^T]]^{-1} = RW^T(WRW^T + Q)^{-1}y$$

和

$$\begin{aligned}
E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] &= E[\beta\beta^T] - E[\beta y^T][E[yy^T]]^{-1}E[y\beta^T] \\
&= R - RW^T(WRW^T + Q)^{-1}WR^T \\
&= R - RW^T(WRW^T + Q)^{-1}WR,
\end{aligned}$$

自 $E[y\beta^T] = WR^T = WR$ 起。 □

Corollary 5.2. *The estimate given by corollary 5.1 can be expressed in the alternative form*

$$\hat{\beta} = (W^T Q^{-1} W + R^{-1})^{-1} W^T Q^{-1} y \quad (5)$$

with corresponding error covariance

$$E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] = (W^T Q^{-1} W + R^{-1})^{-1}.$$

Proof. 我们需要证明这一点 $RW^T(WRW^T + Q)^{-1} = (W^T Q^{-1} W + R^{-1})^{-1} W^T Q^{-1}$ 。我们通过将两边前乘以 $(WRW^T + Q)$ 并在两边后乘以 $(W^T Q^{-1} W + R^{-1})$ 来证明这一点。

$$\begin{aligned}
&(W^T Q^{-1} W + R^{-1})[RW^T(WRW^T + Q)^{-1}](WRW^T + Q) = \\
&(W^T Q^{-1} W + R^{-1})[(W^T Q^{-1} W + R^{-1})^{-1} W^T Q^{-1}](WRW^T + Q) \\
&\Leftrightarrow (W^T Q^{-1} W + R^{-1})RW^T = W^T Q^{-1}(WRW^T + Q) \\
&\Leftrightarrow W^T Q^{-1} WRW^T + \underbrace{R^{-1} R}_I W^T = W^T Q^{-1} WRW^T + W^T \underbrace{Q^{-1} Q}_I \\
&\Leftrightarrow W^T Q^{-1} WRW + W^T = W^T Q^{-1} WRW^T + W^T \checkmark
\end{aligned}$$

Substituting in corollary 5.1 we have

$$\begin{aligned}
E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= R - RW^T(WRW^T + Q)^{-1}WR \\
&= R - (W^TQ^{-1}W + R^{-1})^{-1}W^TQ^{-1}WR \\
&= (W^TQ^{-1}W + R^{-1})^{-1}(W^TQ^{-1}W + R^{-1})R \\
&\quad - (W^TQ^{-1}W + R^{-1})^{-1}W^TQ^{-1}WR \\
&= (W^TQ^{-1}W + R^{-1})^{-1}[(W^TQ^{-1}W + R^{-1})R - W^TQ^{-1}WR] \\
&= (W^TQ^{-1}W + R^{-1})^{-1}[W^TQ^{-1}WR + R^{-1}R - W^TQ^{-1}WR] \\
&= (W^TQ^{-1}W + R^{-1})^{-1} \checkmark
\end{aligned}$$

□

If we compare equation 5 of corollary 5.1 to the Gauss-Markov estimate we see that if $R^{-1} = 0$, corresponding to infinite variance of prior on β , then the minimum-variance estimate is equal to the Gauss-Markov estimate. In other words, the Gauss-Markov estimate is a special case of the minimum-variance estimate, when we have no prior information on β .

5.2 Preliminary Theorems

Theorem 5.2. *The minimum variance linear estimate of a linear function of β is equal to the linear function of the minimum variance estimate of β . In other words given a matrix T (the linear function), the minimum variance estimate of $T\beta$ is $T\hat{\beta} = TE[\beta y^T][E[yy^T]]^{-1}y$.*

Proof. Using the proof of the minimum variance theorem and by replacing β with $T\beta$ we write

$$\begin{aligned}
E[yy^T]K^T &= E[y(T\beta)^T] = E[y\beta^T T^T] = E[y\beta^T]T^T \\
&\Rightarrow K^T = (E[yy^T])^{-1} E[y\beta^T]T^T \\
&\Rightarrow K = T \underbrace{E[\beta y^T](E[yy^T])^{-1}}_{\hat{\beta}} = T\hat{\beta}
\end{aligned}$$

□

Theorem 5.3. *If $\hat{\beta} = Ky$ is the minimum variance estimate of β , then $\hat{\beta}$ is also the linear estimate that minimizes $E[(\beta - \hat{\beta})^T P(\beta - \hat{\beta})]$ for any positive semi-definite $P_{n \times n}$.*

代入推论 5.1 我们有

$$\begin{aligned}
E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= R - RW^T(WRW^T + Q)^{-1}WR \\
&= R - (W^TQ^{-1}W + R^{-1})^{-1}W^TQ^{-1}WR \\
&= (W^TQ^{-1}W + R^{-1})^{-1}(W^TQ^{-1}W + R^{-1})R \\
&\quad - (W^TQ^{-1}W + R^{-1})^{-1}W^TQ^{-1}WR \\
&= (W^TQ^{-1}W + R^{-1})^{-1}[(W^TQ^{-1}W + R^{-1})R - W^TQ^{-1}WR] \\
&= (W^TQ^{-1}W + R^{-1})^{-1}[W^TQ^{-1}WR + R^{-1}R - W^TQ^{-1}WR] \\
&= (W^TQ^{-1}W + R^{-1})^{-1} \checkmark
\end{aligned}$$

□

如果我们将推论 5.1 的方程 5 与高斯-马尔可夫估计进行比较，我们会发现，如果 $R^{-1} = 0$ （对应于 β 上先验的无限方差），则最小方差估计等于高斯-马尔可夫估计。换句话说，当我们没有关于 β 的先验信息时，高斯马尔可夫估计是最小方差估计的特殊情况。

5.2 Preliminary Theorems

Theorem 5.2. *The minimum variance linear estimate of a linear function of β is equal to the linear function of the minimum variance estimate of β . In other words given a matrix T (the linear function), the minimum variance estimate of $T\beta$ is $T\hat{\beta} = TE[\beta y^T][E[yy^T]]^{-1}y$.*

Proof. 使用最小方差定理的证明并用 $T\beta$ 替换 β 我们写

$$\begin{aligned}
E[yy^T]K^T &= E[y(T\beta)^T] = E[y\beta^T T^T] = E[y\beta^T]T^T \\
&\Rightarrow K^T = (E[yy^T])^{-1} E[y\beta^T]T^T \\
&\Rightarrow K = T \underbrace{E[\beta y^T](E[yy^T])^{-1}}_{\hat{\beta}} = T\hat{\beta}
\end{aligned}$$

□

Theorem 5.3. *If $\hat{\beta} = Ky$ is the minimum variance estimate of β , then $\hat{\beta}$ is also the linear estimate that minimizes $E[(\beta - \hat{\beta})^T P(\beta - \hat{\beta})]$ for any positive semi-definite $P_{n \times n}$.*

Proof. Let $P^{\frac{1}{2}}$ be positive square root of P . According to theorem 5.2, $P^{\frac{1}{2}}\hat{\beta}$ is the minimum variance estimate of $P^{\frac{1}{2}}\beta$, which means that $P^{\frac{1}{2}}\hat{\beta}$ minimizes

$$\begin{aligned} E[\|P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta\|_2^2] &= E[(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)^T(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)] \\ &= E[(\hat{\beta}^T P^{\frac{1}{2}T} - \beta^T P^{\frac{1}{2}T})(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)]. \end{aligned}$$

Since $P^{\frac{1}{2}} \geq 0$ then $P^{\frac{1}{2}T} = P^{\frac{1}{2}}$ and the proof follows by noting that

$$\begin{aligned} E[\|P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta\|_2^2] &= E[(\hat{\beta}^T P^{\frac{1}{2}} - \beta^T P^{\frac{1}{2}})(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)] \\ &= E[\hat{\beta}^T P\hat{\beta} - \hat{\beta}^T P\beta - \beta^T P\hat{\beta} + \beta^T P\beta] = * \\ E[(\beta - \hat{\beta})^T P(\beta - \hat{\beta})] &= E[(\beta^T - \hat{\beta}^T)P(\beta - \hat{\beta})] = E[(\beta^T P - \hat{\beta}^T P)(\beta - \hat{\beta})] \\ &= E[\hat{\beta}^T P\hat{\beta} - \hat{\beta}^T P\beta - \beta^T P\hat{\beta} + \beta^T P\beta] = * \end{aligned}$$

□

5.3 Updating the Estimate

We consider the problem of updating the estimate of β if additional data becomes available.

First we define the sum of two vector subspaces $\mathcal{Y}_1 + \mathcal{Y}_2$ of a Hilbert space $\mathcal{H}$ as consisting of all vectors in the form of $y_1 + y_2$ where $y_1 \in \mathcal{Y}_1$ and $y_2 \in \mathcal{Y}_2$. We also define the vector space $\mathcal{Y}$ as the direct sum of two vector subspaces $\mathcal{Y} = \mathcal{Y}_1 \oplus \mathcal{Y}_2$ if every vector $y \in \mathcal{Y}$ has a unique representation in the form of $y = y_1 + y_2$ where $y_1 \in \mathcal{Y}_1$ and $y_2 \in \mathcal{Y}_2$.

We know that if $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are two subspaces of a Hilbert space $\mathcal{H}$, then $\mathcal{Y}_1 + \mathcal{Y}_2$ is also a subspace of the space. We also know that if the subspace $\mathcal{Y}_2$ is chosen such that $\mathcal{Y}_2 \perp \mathcal{Y}_1$ and $\mathcal{Y}_2 \oplus \mathcal{Y}_1 = \mathcal{Y}_1 + \mathcal{Y}_2^1$, then the projection of a vector $\beta \in \mathcal{H}$ onto $\mathcal{Y}_1 + \mathcal{Y}_2$ is equal to the projection of β onto $\mathcal{Y}_1$ plus the projection of β onto $\mathcal{Y}_2$. This is visualized in Figure 1.

Theorem 5.4. *Let β_i be a random variable and $\hat{\beta}_{i1}$ be the minimum variance estimate of β_i , given the random vector y_1 . Just like the proof of the minimum variance estimator theorem, the elements of y_1 span a subspace $\mathcal{Y}_1 \triangleq$ all linear combinations of the elements of y_1 .*

Let y_2 be a random vector and the elements of y_2 span a subspace $\mathcal{Y}_2$. Let $\hat{y}_2$ be the minimum variance estimate of y_2 in $\mathcal{Y}_1$. By the minimum variance estimate theorem, this is equivalent to saying that $\hat{y}_2$ is the orthogonal projection of the elements of y_2 onto $\mathcal{Y}_1$.

¹meaning that $\mathcal{Y}_2 \oplus \mathcal{Y}_1$ produces a subspace that is equal to the subspace of $\mathcal{Y}_1 + \mathcal{Y}_2$

Proof. 令 $P^{\frac{1}{2}}$ 为 P 的正平方根。根据定理 5.2, $P^{\frac{1}{2}}\hat{\beta}$ 是 $P^{\frac{1}{2}}\beta$ 的最小方差估计, 这意味着 $P^{\frac{1}{2}}\hat{\beta}$ 最小化

$$\begin{aligned} E[\|P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta\|_2^2] &= E[(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)^T(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)] \\ &= E[(\hat{\beta}^T P^{\frac{1}{2}T} - \beta^T P^{\frac{1}{2}T})(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)]. \end{aligned}$$

由于 $P^{\frac{1}{2}} \geq 0$ 那么 $P^{\frac{1}{2}T} = P^{\frac{1}{2}}$ 并且证明如下:

$$\begin{aligned} E[\|P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta\|_2^2] &= E[(\hat{\beta}^T P^{\frac{1}{2}} - \beta^T P^{\frac{1}{2}})(P^{\frac{1}{2}}\hat{\beta} - P^{\frac{1}{2}}\beta)] \\ &= E[\hat{\beta}^T P\hat{\beta} - \hat{\beta}^T P\beta - \beta^T P\hat{\beta} + \beta^T P\beta] = * \\ E[(\beta - \hat{\beta})^T P(\beta - \hat{\beta})] &= E[(\beta^T - \hat{\beta}^T)P(\beta - \hat{\beta})] = E[(\beta^T P - \hat{\beta}^T P)(\beta - \hat{\beta})] \\ &= E[\hat{\beta}^T P\hat{\beta} - \hat{\beta}^T P\beta - \beta^T P\hat{\beta} + \beta^T P\beta] = * \end{aligned}$$

□

5.3 Updating the Estimate

如果有额外的数据可用, 我们考虑更新 β 估计的问题。

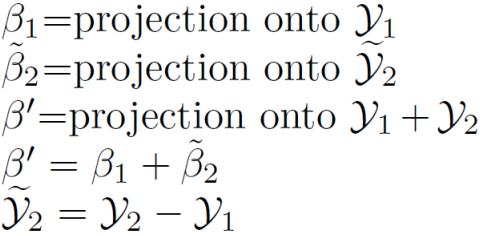
首先, 我们将希尔伯特空间 $\mathcal{H}$ 的两个向量空间 $\mathcal{Y}_1 + \mathcal{Y}_2$ 的和定义为由 $y_1 + y_2$ 形式的所有向量组成, 其中 $y_1 \in \mathcal{Y}_1$ 和 $y_2 \in \mathcal{Y}_2$ 。如果每个向量 $y \in \mathcal{Y}$ 都具有 $y = y_1 + y_2$ 形式的唯一表示 (其中 $y_1 \in \mathcal{Y}_1$ 和 $y_2 \in \mathcal{Y}_2$) , 则我们还将向量空间 $\mathcal{Y}$ 定义为两个向量空间 $\mathcal{Y} = \mathcal{Y}_1 \oplus \mathcal{Y}_2$ 的直接和。

我们知道, 如果 $\mathcal{Y}_1$ 和 $\mathcal{Y}_2$ 是希尔伯特空间 $\mathcal{H}$ 的两个子空间, 那么 $\mathcal{Y}_1 + \mathcal{Y}_2$ 也是该空间的子空间。我们还知道, 如果选择子空间 $\tilde{\mathcal{Y}}_2$ 使得 $\tilde{\mathcal{Y}}_2 \perp \mathcal{Y}_1$ 和 $\tilde{\mathcal{Y}}_2 \oplus \mathcal{Y}_1 = \mathcal{Y}_1 + \mathcal{Y}_2$ ¹, 则向量 $\beta \in \mathcal{H}$ 在 $\mathcal{Y}_1 + \mathcal{Y}_2$ 上的投影等于 β 在 $\mathcal{Y}_1$ 上的投影加上 β 在 $\tilde{\mathcal{Y}}_2$ 上的投影。图 1 对此进行了可视化。

Theorem 5.4. *Let β_i be a random variable and $\hat{\beta}_{i1}$ be the minimum variance estimate of β_i , given the random vector y_1 . Just like the proof of the minimum variance estimator theorem, the elements of y_1 span a subspace $\mathcal{Y}_1 \triangleq$ all linear combinations of the elements of y_1 .*

Let y_2 be a random vector and the elements of y_2 span a subspace $\mathcal{Y}_2$. Let $\hat{y}_2$ be the minimum variance estimate of y_2 in $\mathcal{Y}_1$. By the minimum variance estimate theorem, this is equivalent to saying that $\hat{y}_2$ is the orthogonal projection of the elements of y_2 onto $\mathcal{Y}_1$.

¹意味着 $\tilde{\mathcal{Y}}_2 \oplus \mathcal{Y}_1$ 产生的子空间等于 $\mathcal{Y}_1 + \mathcal{Y}_2$ 的子空间



Let $\tilde{y}_2 = y_2 - \hat{y}_2$, then the minimum variance estimate of β_i , given y_1 and y_2 , is denoted by $\hat{\beta}_{i2}$ and can be found as

This is equal to saying that the orthogonal projection of β_i onto $\mathcal{Y}_1 + \mathcal{Y}_2$ is denoted by $\hat{\beta}_{i2}$. In other words $\hat{\beta}_{i2}$ is $\hat{\beta}_{i1}$ plus the minimum variance estimate of β_i given the random vector $\tilde{y}_2$. This is similar to finding the orthogonal projection of $\hat{\beta}_i$ onto $\mathcal{Y}_2$ which is generated by $\tilde{y}_2$.

Also, since $\tilde{\mathcal{Y}}_2 \perp \mathcal{Y}_1$, this orthogonal projection of β_i onto $\tilde{\mathcal{Y}}_2 \oplus \mathcal{Y}_1$ (which we denote as $\hat{\beta}_{i2}$) is equal to the sum of individual projections onto $\mathcal{Y}_1$ (which is $\hat{\beta}_{i1}$) and onto $\tilde{\mathcal{Y}}_2$ (which is $E[\beta_i \tilde{y}_2^T] [E[\tilde{y}_2 \tilde{y}_2^T]]^{-1} \tilde{y}_2$). Therefore

□

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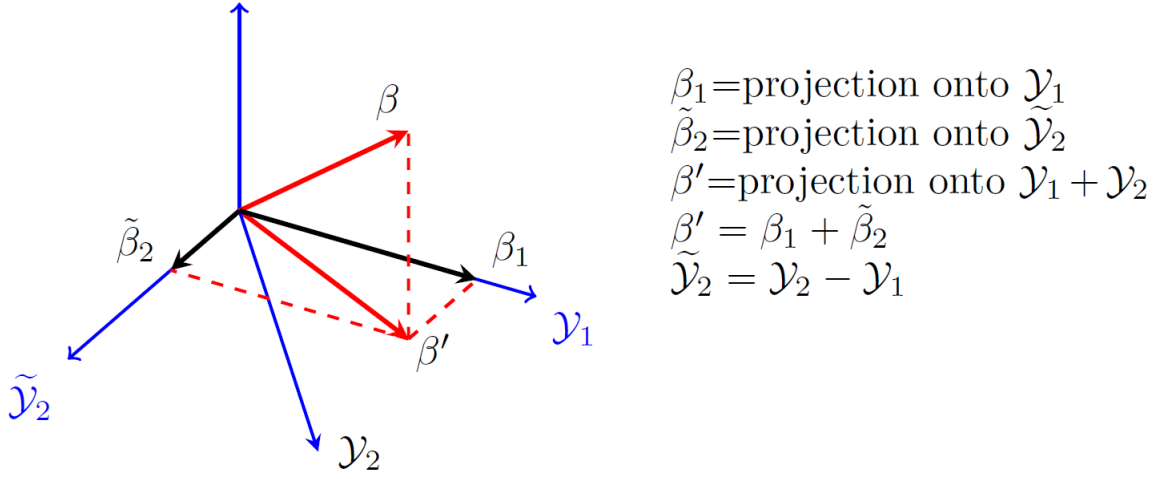


图 1: 向量 β 投影到不同子空间的可视化。

Let $\tilde{y}_2 = y_2 - \hat{y}_2$, then the minimum variance estimate of β_i , given y_1 and y_2 , is denoted by $\hat{\beta}_{i2}$ and can be found as

$$\hat{\beta}_{i2} = \hat{\beta}_{i1} + E[\beta_i \tilde{y}_2^T] [E[\tilde{y}_2 \tilde{y}_2^T]]^{-1} \tilde{y}_2.$$

This is equal to saying that the orthogonal projection of β_i onto $\mathcal{Y}_1 + \mathcal{Y}_2$ is denoted by β_{i2} . In other words β_{i2} is β_{i1} plus the minimum variance estimate of β_i given the random vector $\tilde{y}_2$. This is similar to finding the orthogonal projection of β_i onto $\mathcal{Y}_2$ which is generated by $\tilde{y}_2$.

Proof. 从 $\mathcal{Y}_1 + \mathcal{Y}_2 = \tilde{\mathcal{Y}}_2 \oplus \mathcal{Y}_1$ 开始, β_i 到 $\mathcal{Y}_1 + \mathcal{Y}_2$ 的正交投影与 β_i 到 $\tilde{\mathcal{Y}}_2 \oplus \mathcal{Y}_1$ 的正交投影相同, 如图 1 所示。

此外, 从 $\tilde{\mathcal{Y}}_2 \perp \mathcal{Y}_1$ 开始, β_i 到 $\tilde{\mathcal{Y}}_2 \oplus \mathcal{Y}_1$ (我们表示为 $\hat{\beta}_{i2}$) 的正交投影等于 $\mathcal{Y}_1$ (即 $\hat{\beta}_{i1}$) 和 $\tilde{\mathcal{Y}}_2$ (即 $E[\beta_i \tilde{y}_2^T] [E[\tilde{y}_2 \tilde{y}_2^T]]^{-1} \tilde{y}_2$) 上的各个投影之和。所以

$$\hat{\beta}_{i2} = \hat{\beta}_{i1} + E[\beta_i \tilde{y}_2^T] [E[\tilde{y}_2 \tilde{y}_2^T]]^{-1} \tilde{y}_2.$$

□

Intuition: 给定新数据, 更新基于新数据中与旧数据正交的部分。这意味着更新基于与旧数据 $\mathcal{Y}_1$ 正交的 $\tilde{\mathcal{Y}}_2$ 。

5.3.1 Example on Updating the Estimate

Suppose that an optimal estimate $\hat{\beta}$ of a random vector β has been formed on the basis of past measurements and that

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = R.$$

Given additional measurements of the form $y = W\beta + \varepsilon$, where ε is a random vector of zero mean which is uncorrelated to both β and the past measurements, we seek to find the updated optimal estimate $\hat{\hat{\beta}}$ and the error covariance $E[(\beta - \hat{\hat{\beta}})(\beta - \hat{\hat{\beta}})^T]$.

Using the previous theorem we know that

$$\hat{\hat{\beta}} = \hat{\beta} + E[\beta \tilde{y}^T] [E[\tilde{y} \tilde{y}^T]]^{-1} \tilde{y},$$

where $\tilde{y} = y - \hat{y}$ and $\hat{y}$ is the minimum variance estimate of y given previous measurements. But, the minimum variance estimate of y is equal to the minimum variance estimate of $W\beta$ (since $y = W\beta$), which by Theorem 5.2 is equal to $W\hat{\beta}$. Hence, we have $\tilde{y} = y - W\hat{\beta}$.

Note that $y = W\beta + \varepsilon$ and not $y = W\beta$ but since ε is zero mean and uncorrelated to β and the past measurements, the proofs of Theorems 5.1, 5.2 and Corollary 5.1 makes it clear that the minimum variance estimate of $y = W\beta + \varepsilon$ is $W\hat{\beta}$ since $E[\varepsilon \beta^T] = 0$.

In order to compute $\hat{\hat{\beta}}$ we need to compute $E[\beta \tilde{y}^T]$ and $E[\tilde{y} \tilde{y}^T]$. To do this we must consider a few things regarding $\tilde{y} = y - W\hat{\beta}$.

- y = the new measurement = $W\beta + \varepsilon$.
- $W\hat{\beta}$ = the best estimate of the new measurement y based on the past measurement y_p .
- y_p = the past measurement from which the estimate $\hat{\beta}$ was made. (y_p was used to find $\hat{\beta}$.)
- y_p was also found from the $W\beta + \varepsilon$ process so $y_p = W\beta + \varepsilon$.

We have previously proven that

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = R - RW^T(WRW^T + Q)^{-1}WR \triangleq \mathfrak{R}$$

5.3.1 Example on Updating the Estimate

假设已经根据过去的测量形成了随机向量 β 的最优估计 $\hat{\beta}$ ，并且

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = R.$$

给定 $y = W\beta + \varepsilon$ 形式的附加测量，其中 ε 是零均值的随机向量，与 β 和过去的测量都不相关，我们寻求找到更新的最优估计 $\hat{\beta}$ 和误差协方差 $E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T]$ 。

利用前面的定理我们知道

$$\hat{\beta} = \hat{\beta} + E[\beta \tilde{y}^T] [E[\tilde{y} \tilde{y}^T]]^{-1} \tilde{y},$$

其中 $\tilde{y} = y - \hat{y}$ 和 $\hat{y}$ 是给定先前测量值的 y 的最小方差估计。但是， y 的最小方差估计等于自 $y = W\beta$ 以来 $W\beta$ 的最小方差估计，根据定理 5.2，它等于 $W\hat{\beta}$ 。因此，我们有 $\tilde{y} = y - W\hat{\beta}$ 。

请注意， $y = W\beta + \varepsilon$ 而不是 $y = W\beta$ ，但由于 ε 为零均值且与 β 和过去的测量值不相关，定理 5.1, 5.2 的证明和推论 5.1 清楚地表明，由于 $E[\varepsilon \beta^T] = 0$ ， $y = W\beta + \varepsilon$ 的最小方差估计为 $W\hat{\beta}$ 。

为了计算 $\hat{\beta}$ ，我们需要计算 $E[\beta \tilde{y}^T]$ 和 $E[\tilde{y} \tilde{y}^T]$ 。为此，我们必须考虑有关 $\tilde{y} = y - W\hat{\beta}$ 的一些事情。

- y = 新的测量值 $= W\beta + \varepsilon$ 。
- $W\hat{\beta}$ = 基于过去测量值 y_p 的新测量值 y 的最佳估计。
- y_p = 进行估计 $\hat{\beta}$ 的过去测量值。（ y_p 用于查找 $\hat{\beta}$ 。）
- y_p 也从 $W\beta + \varepsilon$ 进程中找到了 $y_p = W\beta + \varepsilon$ 。

我们之前已经证明了

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = R - RW^T(WRW^T + Q)^{-1}WR \triangleq \mathfrak{R}$$

where

$$\begin{aligned}
Q &= E[\varepsilon\varepsilon^T] \\
R &= E[\beta\beta^T] \\
E[y_p\beta^T] &= WR \\
E[\beta y_p^T] &= RW^T \\
\hat{\beta} &= RW^T(WRW^T + Q)^{-1}y_p \\
\hat{y} &= W\hat{\beta} = WRW^T(WRW^T + Q)^{-1}y_p.
\end{aligned}$$

Using these previous results on the past measurement y_p and by multiplying both sides of the above by W^T from the right and W from the left we have

$$W\Re W^T = WRW^T - WRW^T(WRW^T + Q)^{-1}WRW^T.$$

We can now use the previous formulas to find $E[\beta\tilde{y}^T]$ and $E[\tilde{y}\tilde{y}^T]$ as

$$\begin{aligned}
E[\beta\tilde{y}^T] &= E[\beta(y - W\hat{\beta})^T] &&= E[\beta(y^T - \hat{\beta}^T W^T)] \\
&= E[\beta y^T] - E[\beta\hat{\beta}^T W^T] = E[\beta y^T] - E[\beta\hat{\beta}^T]W^T \\
&= E[\beta(W\beta + \varepsilon)^T] &&- E[\beta y_p^T(WRW^T + Q)^{-1}WR]W^T \\
&= E[\beta(\beta^T W^T + \varepsilon^T)] &&- E[\beta y_p^T](WRW^T + Q)^{-1}WRW^T \\
&= E[\beta\beta^T]W^T + E[\beta\varepsilon^T] - E[\beta y_p^T](WRW^T + Q)^{-1}WRW^T \\
&= RW^T &&+ 0 &&- RW^T(WRW^T + Q)^{-1}WRW^T \\
&= \Re W^T
\end{aligned}$$

and

$$\begin{aligned}
E[\tilde{y}\tilde{y}^T] &= E[(y - W\hat{\beta})(y - W\hat{\beta})^T] = E[(y - W\hat{\beta})(y^T - \hat{\beta}^T W^T)] \\
&= E[yy^T - y\hat{\beta}^T W^T - W\hat{\beta}y + W\hat{\beta}\hat{\beta}^T W^T] \\
&= E[yy^T] - E[y\hat{\beta}^T]W^T - WE[\hat{\beta}y^T] + WE[\hat{\beta}\hat{\beta}^T]W^T \\
&= E[(W\beta + \varepsilon)(W\beta + \varepsilon)^T] - E[(W\beta + \varepsilon)(y_p^T(WRW^T + Q)^{-1}WR)]W^T \\
&\quad - WE[(RW^T(WRW^T + Q)^{-1}y_p)(W\beta + \varepsilon)^T] \\
&\quad + WE[(RW^T(WRW^T + Q)^{-1}y_p)(y_p^T(WRW^T + Q)^{-1}WR)]W^T \\
&= WRW^T + Q - E[(W\beta + \varepsilon)y_p^T](WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}E[y_p(W\beta + \varepsilon)^T] \\
&\quad + WRW^T(WRW^T + Q)^{-1}E[y_p y_p^T](WRW^T + Q)^{-1}WRW^T
\end{aligned}$$

在哪里

$$\begin{aligned}
Q &= E[\varepsilon\varepsilon^T] \\
R &= E[\beta\beta^T] \\
E[y_p\beta^T] &= WR \\
E[\beta y_p^T] &= RW^T \\
\hat{\beta} &= RW^T(WRW^T + Q)^{-1}y_p \\
\hat{y} &= W\hat{\beta} = WRW^T(WRW^T + Q)^{-1}y_p.
\end{aligned}$$

将这些先前的结果用于过去的测量 y_p ，并将上面的两边乘以右侧的 W^T 和左侧的 W ，我们得到

$$W\Re W^T = WRW^T - WRW^T(WRW^T + Q)^{-1}WRW^T.$$

我们现在可以使用前面的公式来查找 $E[\beta\tilde{y}^T]$ 和 $E[\tilde{y}\tilde{y}^T]$ 作为

$$\begin{aligned}
E[\beta\tilde{y}^T] &= E[\beta(y - W\hat{\beta})^T] = E[\beta(y^T - \hat{\beta}^T W^T)] \\
&= E[\beta y^T] - E[\beta\hat{\beta}^T W^T] = E[\beta y^T] - E[\beta\hat{\beta}^T]W^T \\
&= E[\beta(W\beta + \varepsilon)^T] - E[\beta y_p^T(WRW^T + Q)^{-1}WR]W^T \\
&= E[\beta(\beta^T W^T + \varepsilon^T)] - E[\beta y_p^T](WRW^T + Q)^{-1}WRW^T \\
&= E[\beta\beta^T]W^T + E[\beta\varepsilon^T] - E[\beta y_p^T](WRW^T + Q)^{-1}WRW^T \\
&= RW^T + 0 - RW^T(WRW^T + Q)^{-1}WRW^T \\
&= \Re W^T
\end{aligned}$$

和

$$\begin{aligned}
E[\tilde{y}\tilde{y}^T] &= E[(y - W\hat{\beta})(y - W\hat{\beta})^T] = E[(y - W\hat{\beta})(y^T - \hat{\beta}^T W^T)] \\
&= E[yy^T - y\hat{\beta}^T W^T - W\hat{\beta}y + W\hat{\beta}\hat{\beta}^T W^T] \\
&= E[yy^T] - E[y\hat{\beta}^T]W^T - WE[\hat{\beta}y^T] + WE[\hat{\beta}\hat{\beta}^T]W^T \\
&= E[(W\beta + \varepsilon)(W\beta + \varepsilon)^T] - E[(W\beta + \varepsilon)(y_p^T(WRW^T + Q)^{-1}WR)]W^T \\
&\quad - WE[(RW^T(WRW^T + Q)^{-1}y_p)(W\beta + \varepsilon)^T] \\
&\quad + WE[(RW^T(WRW^T + Q)^{-1}y_p)(y_p^T(WRW^T + Q)^{-1}WR)]W^T \\
&= WRW^T + Q - E[(W\beta + \varepsilon)y_p^T](WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}E[y_p(W\beta + \varepsilon)^T] \\
&\quad + WRW^T(WRW^T + Q)^{-1}E[y_p y_p^T](WRW^T + Q)^{-1}WRW^T
\end{aligned}$$

$$\begin{aligned}
&= WRW^T + Q \\
&\quad - E[W\beta y_p^T + \varepsilon y_p^T](WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}E[y_p\beta^T W^T + y_p\varepsilon^T] \\
&\quad + WRW^T(WRW^T + Q)^{-1}(WRW^T + Q)(WRW^T + Q)^{-1}WRW^T \\
&= WRW^T + Q \\
&\quad - (E[W\beta y_p^T] + E[\varepsilon y_p^T])(WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}(E[y_p\beta^T W^T] + E[y_p\varepsilon^T]) \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T \\
&= WRW^T + Q \\
&\quad - WE[\beta y_p^T](WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}E[y_p\beta^T]W^T \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T \\
&= WRW^T + Q \\
&\quad - WRW^T(WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}WRW^T \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T \\
&= WRW^T - WRW^T(WRW^T + Q)^{-1}WRW^T + Q \\
&= W\mathfrak{R}W^T + Q.
\end{aligned}$$

Finally, noting that $\mathfrak{R}^T = \mathfrak{R}$, the error covariance is found as

$$\begin{aligned}
&E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] \\
&= E[(\beta - \hat{\beta} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta}))(\beta - \hat{\beta} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta}))^T] \\
&= E[(\beta - \hat{\beta} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta}))((\beta - \hat{\beta})^T - (y - W\hat{\beta})^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R})] \\
&= E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T - (\beta - \hat{\beta})(y - W\hat{\beta})^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} \\
&\quad - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta})(\beta - \hat{\beta})^T \\
&\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta})(y - W\hat{\beta})^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}].
\end{aligned}$$

$$\begin{aligned}
&= WRW^T + Q \\
&\quad - E[W\beta y_p^T + \varepsilon y_p^T](WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}E[y_p\beta^T W^T + y_p\varepsilon^T] \\
&\quad + WRW^T(WRW^T + Q)^{-1}(WRW^T + Q)(WRW^T + Q)^{-1}WRW^T \\
&= WRW^T + Q[W\beta y_p^T] + E[\varepsilon y_p^T](WRW^T + Q)^{-1}WRW^T \\
&\quad - WRW^T(WRW^T + Q)^{-1}(E[y_p\beta^T W^T] + E[y_p\varepsilon^T]) \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T = WRW^T + Q[\beta y_p^T](WRW^T + Q)^{-1}WRW^T - WRW^T(WRW^T + Q)^{-1}E[y_p\beta^T]W^T \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T \\
&= WRW^T + Q \\
&\quad - WRW^T(WRW^T + Q)^{-1}WRW^T \\
&\quad + WRW^T(WRW^T + Q)^{-1}WRW^T \\
&= WRW^T - WRW^T(WRW^T + Q)^{-1}WRW^T + Q = W\mathfrak{R}W^T + Q.
\end{aligned}$$

最后，注意 $\mathfrak{R}^T = \mathfrak{R}$ ，误差协方差为

$$\begin{aligned}
E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] &= E[(\beta - \hat{\beta} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta}))(\beta - \hat{\beta} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta}))^T] \\
&= E[(\beta - \hat{\beta} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta}))(\beta - \hat{\beta})^T - (y - W\hat{\beta})^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}] \\
&= E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T - (\beta - \hat{\beta})(y - W\hat{\beta})^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta})(\beta - \hat{\beta})^T \\
&\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(y - W\hat{\beta})(y - W\hat{\beta})^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}].
\end{aligned}$$

Noting that $E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = \mathfrak{R}$, the error covariance is

$$\begin{aligned} &= \mathfrak{R} - E[(\beta - \hat{\beta})(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} \\ &\quad - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})(\beta - \hat{\beta})^T] \\ &\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}. \end{aligned}$$

Since $E[(y - W\hat{\beta})(y - W\hat{\beta})^T] = W\mathfrak{R}W^T + Q$, the error covariance is

$$\begin{aligned} &= \mathfrak{R} - E[\beta(y - W\hat{\beta})^T - \hat{\beta}(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} \\ &\quad - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})\beta^T - (y - W\hat{\beta})\hat{\beta}^T] \\ &\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(W\mathfrak{R}W^T + Q)(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}. \end{aligned}$$

Setting $E[\beta(y - W\hat{\beta})^T] = \mathfrak{R}W^T$ and $E[(y - W\hat{\beta})\beta^T] = W\mathfrak{R}$, the error covariance is

$$\begin{aligned} &= \mathfrak{R} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} + E[\hat{\beta}(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} \\ &\quad - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})\hat{\beta}^T] \\ &\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}. \end{aligned}$$

Finally, if $E[\hat{\beta}(y - W\hat{\beta})^T] = 0$ then the error covariance is

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = \mathfrak{R} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}.$$

To show that $E[\hat{\beta}(y - W\hat{\beta})^T] = 0$ we write

$$\begin{aligned} E[\hat{\beta}(y - W\hat{\beta})^T] &= E[\hat{\beta}y^T - \hat{\beta}\hat{\beta}^TW^T] = E[\hat{\beta}y^T] - E[\hat{\beta}\hat{\beta}^TW^T] \\ &= E[\hat{\beta}(W\beta + \varepsilon)^T] - E[\hat{\beta}\hat{\beta}^TW^T] \\ &= E[\hat{\beta}\beta^T]W^T + E[\hat{\beta}\varepsilon^T] - E[\hat{\beta}\hat{\beta}^TW^T] \\ &= E[Ky_p\beta^T]W^T + E[Ky_p\varepsilon^T] - E[Ky_py_p^TK^T]W^T \\ &= KE[y_p\beta^T]W^T + KE[y_p\varepsilon^T] - KE[y_py_p^T]K^TW^T. \end{aligned}$$

Setting $E[y_p\varepsilon^T] = 0$ and $K = E[\beta y_p^T][E[y_py_p^T]]^{-1}$, the above equation is

$$\begin{aligned} &= E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T]W^T - E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_py_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T]W^T \\ &= E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T]W^T - E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T]W^T = 0 \end{aligned}$$

注意到 $E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = \mathfrak{R}$, 误差协方差为

$$\begin{aligned} &= \mathfrak{R} - E[(\beta - \hat{\beta})(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} \\ &\quad - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})(\beta - \hat{\beta})^T] \\ &\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}. \end{aligned}$$

由于 $E[(y - W\hat{\beta})(y - W\hat{\beta})^T] = W\mathfrak{R}W^T + Q$, 误差协方差为

$$\begin{aligned} &= \mathfrak{R} - E[\beta(y - W\hat{\beta})^T - \hat{\beta}(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} \\ &\quad - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})\beta^T - (y - W\hat{\beta})\hat{\beta}^T] \\ &\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}(W\mathfrak{R}W^T + Q)(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}. \end{aligned}$$

设定 $E[\beta(y - W\hat{\beta})^T] = \mathfrak{R}W^T$ 和 $E[(y - W\hat{\beta})\beta^T] = W\mathfrak{R}$, 误差协方差为

$$\begin{aligned} &= \mathfrak{R} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} + E[\hat{\beta}(y - W\hat{\beta})^T](W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} \\ &\quad - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}E[(y - W\hat{\beta})\hat{\beta}^T] \\ &\quad + \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}. \end{aligned}$$

最后, 如果 $E[\hat{\beta}(y - W\hat{\beta})^T] = 0$ 则误差协方差为

$$E[(\beta - \hat{\beta})(\beta - \hat{\beta})^T] = \mathfrak{R} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R}.$$

为了证明 $E[\hat{\beta}(y - W\hat{\beta})^T] = 0$ 我们写

$$\begin{aligned} E[\hat{\beta}(y - W\hat{\beta})^T] &= E[\hat{\beta}y^T - \hat{\beta}\hat{\beta}^TW^T] = E[\hat{\beta}y^T] - E[\hat{\beta}\hat{\beta}^TW^T] \\ &= E[\hat{\beta}(W\beta + \varepsilon)^T] - E[\hat{\beta}\hat{\beta}^T]W^T \\ &= E[\hat{\beta}\beta^T]W^T + E[\hat{\beta}\varepsilon^T] - E[\hat{\beta}\hat{\beta}^T]W^T \\ &= E[Ky_p\beta^T]W^T + E[Ky_p\varepsilon^T] - E[Ky_py_p^TK^T]W^T \\ &= KE[y_p\beta^T]W^T + KE[y_p\varepsilon^T] - KE[y_py_p^T]K^TW^T. \end{aligned}$$

设定 $E[y_p\varepsilon^T] = 0$ 和 $K = E[\beta y_p^T][E[y_py_p^T]]^{-1}$, 上式为

$$\begin{aligned} &= E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T]W^T - E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_py_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T] \\ &\quad]W^T = E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T]W^T - E[\beta y_p^T][E[y_py_p^T]]^{-1}E[y_p\beta^T]W^T = 0 \end{aligned}$$

6 Kalman Filtering

6.1 Dynamic Model of a Random Process

An n -dimensional dynamic random process can be modeled as follows.

1. A vector difference equation

$$x_{k+1} = \Phi_k x_k + u_k, \quad k = 0, 1, 2, \dots$$

which defines how the random vector x_k changes.

- Here x_k is an n -dimensional state vector where each component is a random variable.
- Φ_k is a known $n \times n$ matrix.
- u_k is an n -dimensional input random vector with zero mean such that there is zero correlation between present input at k and past input at l , i.e:

$$E[u_k u_l^T] = Q_k \delta_{kl} = \begin{cases} Q_k & k = l \\ 0 & k \neq l \end{cases}$$

where $Q_k > 0$ is a positive definite matrix.

2. An initial random vector x_0 and initial random estimate $\hat{x}_0$ with initial error covariance $E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T] = P_0$.
3. Measurements of the process in the form of

$$z_k = H_k x_k + w_k, \quad k = 0, 1, 2, \dots$$

which defines how the measurements z_k of the process x_k are recorded.

- Here H_k is a known $m \times n$ matrix.
- w_k is an n -dimensional random measurement error with zero mean and

$$E[w_k w_l^T] = R_k \delta_{kl} = \begin{cases} R_k & k = l \\ 0 & k \neq l \end{cases}$$

where $R_k > 0$ is a positive definite matrix.

It is assumed that x_0, u_j, w_k are all uncorrelated for $j \geq 0, k \geq 0$.

6 Kalman Filtering

6.1 Dynamic Model of a Random Process

n 维动态随机过程可以建模如下。

1. 向量差分方程

$$x_{k+1} = \Phi_k x_k + u_k, \quad k = 0, 1, 2, \dots$$

它定义了随机向量 x_k 如何变化。

- 这里 x_k 是一个 n 维状态向量，其中每个分量都是随机变量。
- Φ_k 是已知的 $n \times n$ 矩阵。
- u_k 是一个均值为零的 n 维输入随机向量，使得 k 处的当前输入与 l 处的过去输入之间存在零相关性，即：

$$E[u_k u_l^T] = Q_k \delta_{kl} = \begin{cases} Q_k & k = l \\ 0 & k \neq l \end{cases}$$

其中 $Q_k > 0$ 是正定矩阵。

2. 初始随机向量 x_0 和具有初始误差协方差 $E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T] = P_0$ 的初始随机估计 $\hat{x}_0$ 。

3. 过程测量的形式

$$z_k = H_k x_k + w_k, \quad k = 0, 1, 2, \dots$$

它定义了如何记录进程 x_k 的测量值 z_k 。

- 这里 H_k 是已知的 $m \times n$ 矩阵。
- w_k 是均值为零的 n 维随机测量误差，

$$E[w_k w_l^T] = R_k \delta_{kl} = \begin{cases} R_k & k = l \\ 0 & k \neq l \end{cases}$$

其中 $R_k > 0$ 是正定矩阵。

假设 x_0 、 u_j 、 w_k 对于 $j \geq 0, k \geq 0$ 都是不相关的。

6.2 The Estimation Problem

The estimation problem is defined as finding the minimum-variance estimate of x from measurements z . We say that $\hat{x}_{k|j}$ is the optimal estimate of x_k given j measurements (or observations) of z . In other words, $\hat{x}_{k|j}$ is the projection of x_k onto the space $\mathcal{Z}_j$ generated by the random vectors $z_0, z_1, z_2, \dots, z_j$.

We consider the case of $k \geq j$ of either predicting future or present values given past measurements. Estimating past values is called the smoothing problem, which is substantially the same but with messier equations.

6.3 Kalman Filter Theorem

Theorem 6.1. *The optimal estimate $\hat{x}_{k+1|k}$ of a random state vector can be generated recursively as*

$$\hat{x}_{k+1|k} = \Phi_k \hat{x}_{k|k-1} + \Phi_k P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1}) \quad (6)$$

where P_k is the $n \times n$ error covariance of $\hat{x}_{k|k-1}$ which is itself generated recursively as:

$$P_{k+1} = \Phi_k P_k \{I - H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k\} \Phi_k^T + Q_k \quad (7)$$

The required initial conditions are the initial estimates of $\hat{x}_0 = \hat{x}_{0|-1}$ and its error covariance P_0 .

Proof. Suppose that $z_0, z_1, \dots, z_{k-1}$ have been measured and that the estimate $\hat{x}_{k|k-1}$ and error covariance $P_k = E[(\hat{x}_{k|k-1} - x_k)(\hat{x}_{k|k-1} - x_k)^T]$ have been computed. In other words, we have the projection of x_k onto the subspace $\mathcal{Z}_{k-1}$.

At k , we obtain a new measurement

$$z_k = H_k x_k + w_k$$

which gives us additional information about x_k . This is exactly the situation encountered in the aforementioned example of Section 5.3.1. Specifically, by substituting $\hat{\beta} = \hat{x}_{k|k}$, $\hat{\beta} = \hat{x}_{k|k-1}$, $\mathfrak{R} = P_k$, $W = H_k$, $Q = R_k$, $y = z_k$ in the previous example, the updated estimate of x_k is

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1})$$

with associated error covariance

$$\begin{aligned} P_{k|k} &= P_k - P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k \\ &= P_k \{I - H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k\} \end{aligned}$$

6.2 The Estimation Problem

估计问题被定义为根据测量值 z 找到 x 的最小方差估计。我们说，在给定 z 的 j 测量值（或观测值）的情况下， $\hat{x}_{k|j}$ 是 x_k 的最佳估计。换句话说， $\hat{x}_{k|j}$ 是 x_k 在由随机向量 $z_0, z_1, z_2, \dots, z_j$ 生成的空间 $\mathcal{Z}_j$ 上的投影。

我们考虑在给定过去测量的情况下预测未来值或当前值的 $k \geq j$ 情况。估计过去的值称为平滑问题，其基本相同，但方程更为混乱。

6.3 Kalman Filter Theorem

Theorem 6.1. *The optimal estimate $\hat{x}_{k+1|k}$ of a random state vector can be generated recursively as*

$$\hat{x}_{k+1|k} = \Phi_k \hat{x}_{k|k-1} + \Phi_k P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1}) \quad (6)$$

where P_k is the $n \times n$ error covariance of $\hat{x}_{k|k-1}$ which is itself generated recursively as:

$$P_{k+1} = \Phi_k P_k \{I - H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k\} \Phi_k^T + Q_k \quad (7)$$

The required initial conditions are the initial estimates of $\hat{x}_0 = \hat{x}_{0|-1}$ and its error covariance P_0 .

Proof. 假设已测量 $z_0, z_1, \dots, z_{k-1}$ ，并计算了估计值 $\hat{x}_{k|k-1}$ 和误差协方差 $P_k = E[(\hat{x}_{k|k-1} - x_k)(\hat{x}_{k|k-1} - x_k)^T]$ 。换句话说，我们有 x_k 到子空间 $\mathcal{Z}_{k-1}$ 的投影。

在 k 处，我们获得了新的测量值

$$z_k = H_k x_k + w_k$$

这为我们提供了有关 x_k 的更多信息。这正是前面提到的 5.3.1 节例子中遇到的情况。具体来说，通过替换前面示例中的 $\hat{\beta} = \hat{x}_{k|k}$ 、 $\hat{\beta} = \hat{x}_{k|k-1}$ 、 $\mathfrak{R} = P_k$ 、 $W = H_k$ 、 $Q = R_k$ 、 $y = z_k$ ， x_k 的更新估计为

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1})$$

以及相关的误差协方差

$$\begin{aligned} P_{k|k} &= P_k - P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k \\ &= P_k \{I - H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k\} \end{aligned}$$

where $\mathfrak{R} = P_k$ and $\mathfrak{R} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} = P_{k|k}$.

Based on this optimal estimate of x_k , we can compute the optimal estimate $\hat{x}_{k+1|k}$ of $x_{k+1} = \Phi_k \hat{x}_k + u_k$. We can do this using Theorem 5.2, which says that the optimal estimate of $\Phi_k x_k$ is $\Phi_k \hat{x}_{k|k}$, and since u_k is uncorrelated with z_k and x_k , the optimal estimate of x_{k+1} is

$$\hat{x}_{k+1|k} = \Phi_k \hat{x}_{k|k} = \Phi_k \hat{x}_{k|k-1} + \Phi_k P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1}).$$

This proves equation 6.

To prove the error covariance update equation 7, we first note that from Theorem 5.2 we have

$$\begin{aligned} E[(T\hat{\beta} - T\beta)(T\hat{\beta} - T\beta)^T] &= E[T(\hat{\beta} - \beta)(T(\hat{\beta} - \beta))^T] \\ &= E[T(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T T^T] = T E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] T^T. \end{aligned}$$

The error covariance update P_{k+1} is now

$$\begin{aligned} P_{k+1} &= \\ &= E[(\hat{x}_{k+1|k} - x_{k+1})(\hat{x}_{k+1|k} - x_{k+1})^T] \\ &= E[(\Phi_k \hat{x}_{k|k} - (\Phi_k x_k + u_k))(\Phi_k \hat{x}_{k|k} - (\Phi_k x_k + u_k))^T] \\ &= E[(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)^T] \\ &\quad - E[(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)u_k^T] - E[u_k(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)^T] \\ &\quad + E[u_k u_k^T]. \end{aligned}$$

Since the error u_k is uncorrelated with previous estimates

$$E[(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)u_k^T] = E[u_k(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)^T] = 0.$$

Also, we know that $E[u_k u_k^T] = Q_k$, therefore

$$\begin{aligned} P_{k+1} &= \Phi_k E[(\hat{x}_{k|k} - x_k)(\hat{x}_{k|k} - x_k)^T] \Phi_k^T + Q_k \\ &= \Phi_k P_{k|k} \Phi_k^T + Q_k \\ &= \Phi_k P_k \{I - H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k\} \Phi_k^T + Q_k. \end{aligned}$$

□

其中 $\mathfrak{R} = P_k$ 和 $\mathfrak{R} - \mathfrak{R}W^T(W\mathfrak{R}W^T + Q)^{-1}W\mathfrak{R} = P_{k|k\circ}$

基于 x_k 的最优估计, 我们可以计算 $x_{k+1} = \Phi_k \hat{x}_k + u_k$ 的最优估计 $\hat{x}_{k+1|k}$ 。我们可以使用定理 5.2 来做到这一点, 它说 $\Phi_k x_k$ 的最佳估计是 $\Phi_k \hat{x}_{k|k}$, 并且由于 u_k 与 z_k 和 x_k 不相关, 因此 x_{k+1} 的最佳估计是

$$\hat{x}_{k+1|k} = \Phi_k \hat{x}_{k|k} = \Phi_k \hat{x}_{k|k-1} + \Phi_k P_k H_k^T [H_k P_k H_k^T + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1}).$$

这就证明了方程6。

为了证明误差协方差更新方程 7, 我们首先注意到, 根据定理 5.2, 我们有

$$\begin{aligned} E[(T\hat{\beta} - T\beta)(T\hat{\beta} - T\beta)^T] &= E[T(\hat{\beta} - \beta)(T(\hat{\beta} - \beta))^T] \\ &= E[T(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T T^T] = T E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] T^T. \end{aligned}$$

现在误差协方差更新 P_{k+1}

$$\begin{aligned} P_{k+1} &= \\ &= E[(\hat{x}_{k+1|k} - x_{k+1})(\hat{x}_{k+1|k} - x_{k+1})^T] \\ &= E[(\Phi_k \hat{x}_{k|k} - (\Phi_k x_k + u_k))(\Phi_k \hat{x}_{k|k} - (\Phi_k x_k + u_k))^T] \\ &= E[(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)^T] \\ &\quad - E[(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)u_k^T] - E[u_k(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)^T] \\ &\quad + E[u_k u_k^T]. \end{aligned}$$

由于误差 u_k 与之前的估计不相关

$$E[(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)u_k^T] = E[u_k(\Phi_k \hat{x}_{k|k} - \Phi_k x_k)^T] = 0.$$

另外, 我们知道 $E[u_k u_k^T] = Q_k$, 因此

$$\begin{aligned} P_{k+1} &= \Phi_k E[(\hat{x}_{k|k} - x_k)(\hat{x}_{k|k} - x_k)^T] \Phi_k^T + Q_k \\ &= \Phi_k P_{k|k} \Phi_k^T + Q_k \\ &= \Phi_k P_k \{I - H_k^T [H_k P_k H_k^T + R_k]^{-1} H_k P_k\} \Phi_k^T + Q_k. \end{aligned}$$

□

Part II

Bayesian Optimal Filtering

7 General Case

From a Bayesian perspective, filtering means to quantify a degree of belief in the state x_k at time k , given all the data up to time k (Z_k) in a recursive (sequential) manner. i.e. to construct the posterior $P(x_k|Z_k)$. We do this in 2 steps:

1. Prediction: uses the state model to predict the belief of state at time k , using Z_{k-1} .
2. Update: At time k when measurement z_k becomes available, we will update the prediction.

In the step of prediction we have a previous belief $P(x_{k-1}|Z_{k-1})$ and we want to know what can be predicted about x_k i.e. we want to find $P(x_k|Z_{k-1})$. We use the Chapman-Kolmogorov equation:

$$\begin{aligned} P(x_k|Z_{k-1}) &= \frac{P(x_k, Z_{k-1})}{P(Z_{k-1})} = \frac{\int p(x_k, x_{k-1}, Z_{k-1}) dx_{k-1}}{P(Z_{k-1})} \\ &= \frac{\int p(x_k|x_{k-1}, Z_{k-1}) p(x_{k-1}|Z_{k-1}) P(Z_{k-1}) dx_{k-1}}{P(Z_{k-1})} \\ &= \int p(x_k|x_{k-1}, Z_{k-1}) p(x_{k-1}|Z_{k-1}) dx_{k-1} \\ &= \int p(x_k|x_{k-1}) p(x_{k-1}|Z_{k-1}) dx_{k-1} \end{aligned}$$

Assuming the state at time k is only dependent on the state at time $k-1$ and is independent of the observation history Z_{k-1} when x_{k-1} is given. In the above equation $p(x_k|x_{k-1})$ is derived from the state equation.

The step of update uses new measurement z_k to construct the posterior $P(x_k|Z_k)$. The update or corrector is carried out via the Bayes rule.

$$\begin{aligned} P(x_k|Z_k) &= P(x_k|z_k, Z_{k-1}) = \frac{P(x_k, z_k|Z_{k-1})}{P(z_k|Z_{k-1})} = \\ &= \frac{P(z_k|x_k, Z_{k-1}) P(x_k|Z_{k-1})}{P(z_k|Z_{k-1})} \end{aligned}$$

Part II

Bayesian Optimal Filtering

7 General Case

从贝叶斯的角度来看，过滤意味着以递归（顺序）方式给定截至时间 k (Z_k) 的所有数据，量化时间 k 时状态 x_k 的置信度。即构造后验 $P(x_k|Z_k)$ 。我们分两步进行：

1. 预测：使用状态模型来预测时间 k 时状态的置信度，使用 Z_{k-1} 。
2. 更新：在时间 k ，当测量 z_k 可用时，我们将更新预测。

在预测步骤中，我们有一个先前的信念 $P(x_{k-1}|Z_{k-1})$ ，我们想知道关于 x_k 可以预测什么，即我们想找到 $P(x_k|Z_{k-1})$ 。我们使用 Chapman-Kolmogorov 方程：

$$\begin{aligned} P(x_k|Z_{k-1}) &= \frac{P(x_k, Z_{k-1})}{P(Z_{k-1})} = \frac{\int p(x_k, x_{k-1}, Z_{k-1}) dx_{k-1}}{P(Z_{k-1})} \\ &= \frac{\int p(x_k|x_{k-1}, Z_{k-1}) p(x_{k-1}|Z_{k-1}) P(Z_{k-1}) dx_{k-1}}{P(Z_{k-1})} \\ &= \int p(x_k|x_{k-1}, Z_{k-1}) p(x_{k-1}|Z_{k-1}) dx_{k-1} \\ &= \int p(x_k|x_{k-1}) p(x_{k-1}|Z_{k-1}) dx_{k-1} \end{aligned}$$

假设在给定 x_{k-1} 时，时间 k 的状态仅依赖于时间 $k-1$ 的状态，并且与观察历史 Z_{k-1} 无关。上式中的 $p(x_k|x_{k-1})$ 是由状态方程推导出来的。

更新步骤使用新的测量 z_k 来构造后验 $P(x_k|Z_k)$ 。更新或校正通过贝叶斯规则进行的。

$$\begin{aligned} P(x_k|Z_k) &= P(x_k|z_k, Z_{k-1}) = \frac{P(x_k, z_k|Z_{k-1})}{P(z_k|Z_{k-1})} = \\ &= \frac{P(z_k|x_k, Z_{k-1}) P(x_k|Z_{k-1})}{P(z_k|Z_{k-1})} \end{aligned}$$

Assuming that new measurement z_k is independent of the previous measurements Z_{k-1} we may find the update or corrector:

$$P(x_k|Z_k) = \frac{P(z_k|x_k)P(x_k|Z_{k-1})}{P(z_k|Z_{k-1})}$$

$P(z_k|Z_{k-1})$ can be calculated as follows:

$$\begin{aligned} P(z_k|Z_{k-1}) &= \int p(z_k, x_k|Z_{k-1})dx_k = \int p(z_k|x_k, Z_{k-1})p(x_k|Z_{k-1})dx_k \\ &= \int p(z_k|x_k)p(x_k|Z_{k-1})dx_k. \end{aligned}$$

In the above equation $p(z_k|x_k)$ is the likelihood function (likelihood of data z_k given the state x_k) which can be found from the measurement equation.

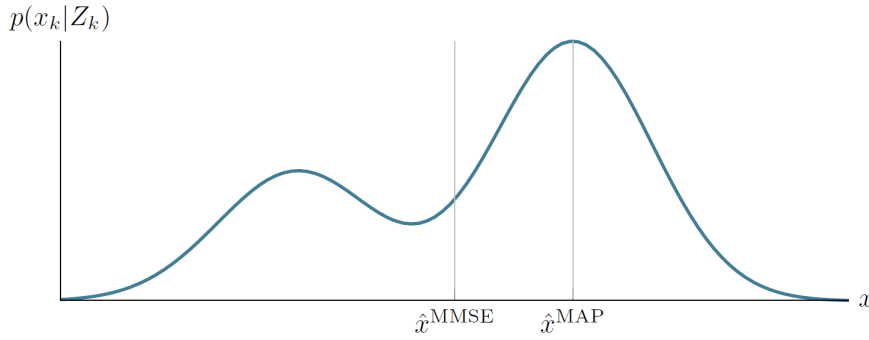
Once the posterior is found, the estimate of the state can be found using the mean or mode of the posterior.

For MMSE this estimate is defined by:

$$\hat{x}_{k|k} = E[x_k|Z_k] = \int x_k p(x_k|Z_k)dx_k \quad (\text{MMSE})$$

For MAP the estimate is given by:

$$\hat{x}_{k|k} = \underset{x_k}{\operatorname{argmax}} \quad p(x_k|Z_k) \quad (\text{MAP})$$



8 Kalman Filtering

The predictor and corrector steps can not be performed analytically for every arbitrary functions ϕ_{k-1} , h_k . However it has a close-form solution for the most simple form of linear/Gaussian (Kalman filter) model.

假设新的测量值 z_k 独立于之前的测量值 Z_{k-1} ，我们可以找到更新或校正器：

$$P(x_k|Z_k) = \frac{P(z_k|x_k)P(x_k|Z_{k-1})}{P(z_k|Z_{k-1})}$$

$P(z_k|Z_{k-1})$ 可以计算如下：

$$\begin{aligned} P(z_k|Z_{k-1}) &= \int p(z_k, x_k|Z_{k-1}) dx_k = \int p(z_k|x_k, Z_{k-1}) p(x_k|Z_{k-1}) dx_k \\ &= \int p(z_k|x_k) p(x_k|Z_{k-1}) dx_k. \end{aligned}$$

上式中 $p(z_k|x_k)$ 是似然函数（给定状态 x_k 的情况下数据 z_k 的似然），可以从测量方程中找到。

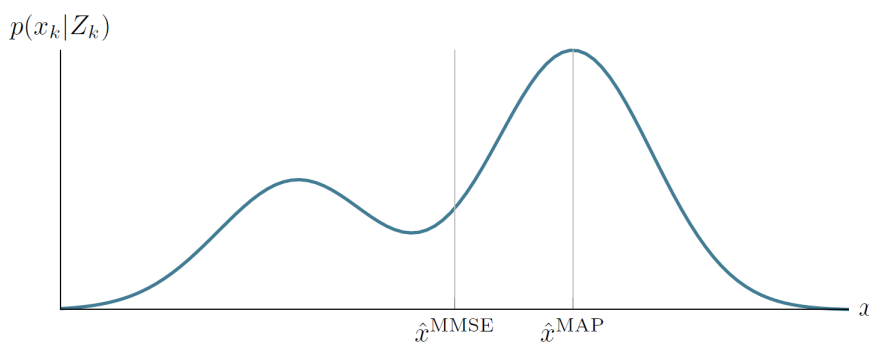
一旦找到后验，就可以使用后验的平均值或众数找到状态的估计。

对于 MMSE，该估计值定义为：

$$\hat{x}_{k|k} = E[x_k|Z_k] = \int x_k p(x_k|Z_k) dx_k \quad (\text{MMSE})$$

对于 MAP，估计值由下式给出：

$$\hat{x}_{k|k} = \underset{x_k}{\operatorname{argmax}} p(x_k|Z_k) \quad (\text{MAP})$$



8 Kalman Filtering

无法对每个任意函数 ϕ_{k-1} 、 h_k 分析执行预测器和校正器步骤。然而，它对于最简单形式的线性/高斯（卡尔曼滤波器）模型有一个近似形式的解决方案。

Sequential Bayesian equations are obtained from the previous discussion

$$\begin{cases} P(x_k|Z_{k-1}) = \int p(x_k|x_{k-1})p(x_{k-1}|Z_{k-1})dx_{k-1} & \text{prediction} \\ P(x_k|Z_k) = \frac{P(z_k|x_k)P(x_k|Z_{k-1})}{\int p(z_k|x_k)p(x_k|Z_{k-1})dx_k} & \text{update/correction} \end{cases}$$

Prediction

In the above equations $p(x_{k-1}|Z_{k-1})$ corresponds to the state x_{k-1} up to time $k-1$, under the Kalman filter settings, this density turns out to be Gaussian

$$x_{k-1}|Z_{k-1} \sim \mathcal{G}(x_{k-1}, \hat{x}_{k-1|k-1}, P_{k-1|k-1})$$

where $\hat{x}_{k-1|k-1}, P_{k-1|k-1}$ can be found from previous steps and going back to x_0 .

$p(x_k|x_{k-1})$ can be found from the state equation

$$p(x_k|x_{k-1}) = \mathcal{G}(x_k, \Phi_{k-1}x_{k-1}, Q_{k-1}) \quad \text{or} \quad x_k|x_{k-1} \sim \mathcal{G}(x_k, \Phi_{k-1}x_{k-1}, Q_{k-1})$$

Now we can put the above equations into the prediction equation

$$\begin{aligned} P(x_k|Z_{k-1}) &= \int \mathcal{G}(x_k, \Phi_{k-1}x_{k-1}, Q_{k-1})\mathcal{G}(x_{k-1}, \hat{x}_{k-1|k-1}, P_{k-1|k-1})dx_{k-1} \\ &= \mathcal{G}(x_k, \Phi_{k-1}\hat{x}_{k-1}, P_{k|k-1}) = \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) \end{aligned}$$

Since the integrand is the multiply of 2 Gaussian PDFs, the result of the integral can be computed in form of a Gaussian PDF with

$$\begin{cases} \hat{x}_{k|k-1} = \Phi_{k-1}\hat{x}_{k-1|k-1} & \text{mean} \\ P_{k|k-1} = \Phi_{k-1}P_{k-1|k-1}\Phi_{k-1}^T + Q_{k-1} & \text{covariance} \end{cases}$$

Note that without new measurements to do the update(correction) step the covariance grows with time.

Correction

In order to find $P(x_k|Z_k)$, we need to compute $P(z_k|x_k)$ and $P(x_k|Z_{k-1})$. $P(z_k|x_k)$ can be found from the measurement equation $z_k = H_kx_k + w_k$ and $P(x_k|Z_{k-1})$ is found in the previous step. We have assumed that $w_k \sim \mathcal{G}(w_k, 0, R_k)$ therefore

$$\begin{aligned} p(z_k|x_k) &= \mathcal{G}(z_k, H_kx_k, R_k) \\ P(x_k|Z_{k-1}) &= \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) \end{aligned}$$

Putting the above PDFs in the corrector we obtain:

$$\begin{aligned} P(x_k|Z_k) &= \frac{P(z_k|x_k)P(x_k|Z_{k-1})}{\int p(z_k|x_k)p(x_k|Z_{k-1})dx_k} \\ P(x_k|Z_k) &= \frac{\mathcal{G}(z_k, H_kx_k, R_k)\mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1})}{\int \mathcal{G}(z_k, H_kx_k, R_k)\mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1})dx_k} \end{aligned}$$

序贯贝叶斯方程由前面的讨论得到

$$\begin{cases} P(x_k|Z_{k-1}) = \int p(x_k|x_{k-1})p(x_{k-1}|Z_{k-1})dx_{k-1} & \text{预言} \\ P(x_k|Z_k) = \frac{P(z_k|x_k)P(x_k|Z_{k-1})}{\int p(z_k|x_k)p(x_k|Z_{k-1})dx_k} & \text{更新/修正} \end{cases}$$

Prediction

在上面的等式中, $p(x_{k-1}|Z_{k-1})$ 对应于时间 $k-1$ 之前的状态 x_{k-1} , 在卡尔曼滤波器设置下, 该密度结果为高斯密度

$$x_{k-1}|Z_{k-1} \sim \mathcal{G}(x_{k-1}, \hat{x}_{k-1|k-1}, P_{k-1|k-1})$$

其中 $\hat{x}_{k-1|k-1}, P_{k-1|k-1}$ 可以从之前的步骤中找到并返回到 x_0 。

$p(x_k|x_{k-1})$ 由状态方程可得

$$p(x_k|x_{k-1}) = \mathcal{G}(x_k, \Phi_{k-1}x_{k-1}, Q_{k-1}) \quad \text{or} \quad x_k|x_{k-1} \sim \mathcal{G}(x_k, \Phi_{k-1}x_{k-1}, Q_{k-1})$$

现在我们可以将上面的方程代入预测方程

$$\begin{aligned} P(x_k|Z_{k-1}) &= \int \mathcal{G}(x_k, \Phi_{k-1}x_{k-1}, Q_{k-1})\mathcal{G}(x_{k-1}, \hat{x}_{k-1|k-1}, P_{k-1|k-1})dx_{k-1} \\ &= \mathcal{G}(x_k, \Phi_{k-1}\hat{x}_{k-1|k-1}, P_{k|k-1}) = \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) \end{aligned}$$

由于被积函数是 2 个高斯 PDF 的乘积, 因此积分结果可以以高斯 PDF 的形式计算:

$$\begin{cases} \hat{x}_{k|k-1} = \Phi_{k-1}\hat{x}_{k-1|k-1} & \text{mean} \\ P_{k|k-1} = \Phi_{k-1}P_{k-1|k-1}\Phi_{k-1}^T + Q_{k-1} & \text{covariance} \end{cases}$$

请注意, 如果没有新的测量来执行更新 (校正) 步骤, 协方差会随着时间而增长。

Correction

为了找到 $P(x_k|Z_k)$, 我们需要计算 $P(z_k|x_k)$ 和 $P(x_k|Z_{k-1})$ 。 $P(z_k|x_k)$ 可以从测量方程 $z_k = H_k x_k + w_k$ 中找到, 而 $P(x_k|Z_{k-1})$ 可以在上一步中找到。我们假设 $w_k \sim \mathcal{G}(w_k, 0, R_k)$ 因此

$$\begin{aligned} p(z_k|x_k) &= \mathcal{G}(z_k, H_k x_k, R_k) \\ P(x_k|Z_{k-1}) &= \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) \end{aligned}$$

将上述 PDF 放入校正器中, 我们得到:

$$\begin{aligned} P(x_k|Z_k) &= \frac{P(z_k|x_k)P(x_k|Z_{k-1})}{\int p(z_k|x_k)p(x_k|Z_{k-1})dx_k} \\ P(x_k|Z_k) &= \frac{\mathcal{G}(z_k, H_k x_k, R_k)\mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1})}{\int \mathcal{G}(z_k, H_k x_k, R_k)\mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1})dx_k} \end{aligned}$$

In order to find the result of the above relation, first we calculate the nominator. It can be shown that (See appendix D of [12]).

$$\det(R_k) \cdot \det(P_{k|k-1}) = \det(R_k + H_k P_{k|k-1} H_k^T)$$

and

$$\begin{aligned} & (z_k - H_k x_k)^T R_k^{-1} (z_k - H_k x_k) + (x_k - \hat{x}_{k|k-1})^T P_{k|k-1}^{-1} (x_k - \hat{x}_{k|k-1}) = \\ & (z_k - H_k \hat{x}_{k|k-1})^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} (z_k - H_k \hat{x}_{k|k-1}) + \\ & (x_k - \hat{x}_{k|k})^T (P_{k|k-1} + H_k^T R_k^{-1} H_k) (x_k - \hat{x}_{k|k}) \end{aligned}$$

Then we write

$$\begin{aligned} & \mathcal{G}(z_k, H_k x_k, R_k) \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) = \\ & \frac{1}{\sqrt{\det(2\pi R_k)}} e^{-\frac{1}{2}(z_k - H_k x_k)^T R_k^{-1} (z_k - H_k x_k)} \frac{1}{\sqrt{\det(2\pi P_{k|k-1})}} e^{-\frac{1}{2}(x_k - \hat{x}_{k|k-1})^T P_{k|k-1}^{-1} (x_k - \hat{x}_{k|k-1})} = \\ & \frac{1}{2\pi \sqrt{\det(R_k) \cdot \det(P_{k|k-1})}} e^{-\frac{1}{2}[(z_k - H_k x_k)^T R_k^{-1} (z_k - H_k x_k) + (x_k - \hat{x}_{k|k-1})^T P_{k|k-1}^{-1} (x_k - \hat{x}_{k|k-1})]} = \\ & \frac{1}{\sqrt{\det(2\pi(R_k + H_k P_{k|k-1} H_k^T))}} e^{-\frac{1}{2}(z_k - H_k \hat{x}_{k|k-1})^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} (z_k - H_k \hat{x}_{k|k-1})} \\ & \times \frac{1}{\sqrt{\det(2\pi(P_{k|k-1} + H_k^T R_k^{-1} H_k))}} e^{-\frac{1}{2}(x_k - \hat{x}_{k|k})^T (P_{k|k-1} + H_k^T R_k^{-1} H_k) (x_k - \hat{x}_{k|k})} \end{aligned}$$

Thus the nominator is

$$\begin{aligned} & \mathcal{G}(z_k, H_k x_k, R_k) \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) = \\ & \mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \mathcal{G}(x_k, \hat{x}_{k|k-1}, (P_{k|k-1} + H_k^T R_k^{-1} H_k)^{-1}) \end{aligned}$$

Secondly integrating the above relationship over x_k gives the denominator

$$\begin{aligned} & \int \mathcal{G}(z_k, H_k x_k, R_k) \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) dx_k = \\ & \mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \underbrace{\int \mathcal{G}(x_k, \hat{x}_{k|k-1}, (P_{k|k-1} + H_k^T R_k^{-1} H_k)^{-1}) dx_k}_1 = \\ & \mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \end{aligned}$$

substituting in the main formula gives the updated PDF. Here the covariance

为了求出上述关系的结果，我们首先计算分子。可以证明（参见[12]的附录D）。

$$\det(R_k) \cdot \det(P_{k|k-1}) = \det(R_k + H_k P_{k|k-1} H_k^T)$$

和

$$\begin{aligned} & (z_k - H_k x_k)^T R_k^{-1} (z_k - H_k x_k) + (x_k - \hat{x}_{k|k-1})^T P_{k|k-1}^{-1} (x_k - \hat{x}_{k|k-1}) = \\ & (z_k - H_k \hat{x}_{k|k-1})^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} (z_k - H_k \hat{x}_{k|k-1}) + \\ & (x_k - \hat{x}_{k|k})^T (P_{k|k-1} + H_k^T R_k^{-1} H_k) (x_k - \hat{x}_{k|k}) \end{aligned}$$

然后我们写

$$\begin{aligned} & \mathcal{G}(z_k, H_k x_k, R_k) \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) = \\ & \frac{1}{\sqrt{\det(2\pi R_k)}} e^{-\frac{1}{2}(z_k - H_k x_k)^T R_k^{-1} (z_k - H_k x_k)} \frac{1}{\sqrt{\det(2\pi P_{k|k-1})}} e^{-\frac{1}{2}(x_k - \hat{x}_{k|k-1})^T P_{k|k-1}^{-1} (x_k - \hat{x}_{k|k-1})} = \\ & \frac{1}{2\pi \sqrt{\det(R_k) \cdot \det(P_{k|k-1})}} e^{-\frac{1}{2}[(z_k - H_k x_k)^T R_k^{-1} (z_k - H_k x_k) + (x_k - \hat{x}_{k|k-1})^T P_{k|k-1}^{-1} (x_k - \hat{x}_{k|k-1})]} = \\ & \frac{1}{\sqrt{\det(2\pi(R_k + H_k P_{k|k-1} H_k^T))}} e^{-\frac{1}{2}(z_k - H_k \hat{x}_{k|k-1})^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} (z_k - H_k \hat{x}_{k|k-1})} \\ & \times \frac{1}{\sqrt{\det(2\pi(P_{k|k-1} + H_k^T R_k^{-1} H_k))}} e^{-\frac{1}{2}(x_k - \hat{x}_{k|k})^T (P_{k|k-1} + H_k^T R_k^{-1} H_k) (x_k - \hat{x}_{k|k})} \end{aligned}$$

因此提名人是

$$\begin{aligned} & \mathcal{G}(z_k, H_k x_k, R_k) \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) = \\ & \mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \mathcal{G}(x_k, \hat{x}_{k|k-1}, (P_{k|k-1} + H_k^T R_k^{-1} H_k)^{-1}) \end{aligned}$$

其次，将上述关系积分到 x_k 上得到分母

$$\begin{aligned} & \int \mathcal{G}(z_k, H_k x_k, R_k) \mathcal{G}(x_k, \hat{x}_{k|k-1}, P_{k|k-1}) dx_k = \\ & \mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \underbrace{\int \mathcal{G}(x_k, \hat{x}_{k|k-1}, (P_{k|k-1} + H_k^T R_k^{-1} H_k)^{-1}) dx_k}_1 = \\ & \mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \end{aligned}$$

代入主公式即可得到更新后的 PDF。这里是协方差

of the updated PDF is defined as $P_{k|k}^{-1} \triangleq P_{k|k-1}^{-1} + H_k^T R_k^{-1} H_k$

$$P(x_k|Z_k) = \frac{\mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \mathcal{G}(x_k, \hat{x}_{k|k}, P_{k|k})}{\underbrace{\mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \int \mathcal{G}(x_k, \hat{x}_{k|k-1}, (P_{k|k-1} + H_k^T R_k^{-1} H_k)^{-1}) dx_k}_1}$$

$$P(x_k|Z_k) = \mathcal{G}(x_k, \hat{x}_{k|k}, P_{k|k})$$

Finally we must find $\hat{x}_{k|k}$ which is the updated estimate. By definition we have

$$P_{k|k}^{-1} \hat{x}_{k|k} = P_{k|k-1}^{-1} \hat{x}_{k|k-1} + H_k^T R_k^{-1} z_k$$

To obtain $\hat{x}_{k|k}$ we find $P_{k|k}$ using the matrix inversion lemma²

$$\begin{aligned} P_{k|k} &= \left[P_{k|k}^{-1} \right]^{-1} = (P_{k|k-1}^{-1} + H_k^T R_k^{-1} H_k)^{-1} \\ &= P_{k|k-1} - P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} H_k P_{k|k-1} \\ &= (I - P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} H_k) P_{k|k-1} \\ &= (I - K_k H_k) P_{k|k-1}, \quad K_k \triangleq P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} \end{aligned}$$

Then by multiplying $P_{k|k}$ and $P_{k|k}^{-1} \hat{x}_{k|k}$ we get

$$\begin{aligned} P_{k|k} P_{k|k}^{-1} \hat{x}_{k|k} &= (I - K_k H_k) P_{k|k-1} \left((P_{k|k-1}^{-1} \hat{x}_{k|k-1} + H_k^T R_k^{-1} z_k) \right) \\ \hat{x}_{k|k} &= (I - K_k H_k) \hat{x}_{k|k-1} + (I - K_k H_k) P_{k|k-1} H_k^T R_k^{-1} z_k \\ &= \hat{x}_{k|k-1} - K_k H_k \hat{x}_{k|k-1} + P_{k|k-1} H_k^T R_k^{-1} z_k - K_k H_k P_{k|k-1} H_k^T R_k^{-1} z_k \\ &= \hat{x}_{k|k-1} + (P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} (R_k + H_k P_{k|k-1} H_k^T) R_k^{-1} \\ &\quad - K_k H_k P_{k|k-1} H_k^T R_k^{-1}) z_k - K_k H_k \hat{x}_{k|k-1} \\ &= \hat{x}_{k|k-1} + (K_k (I + H_k P_{k|k-1} H_k^T R_k^{-1}) - K_k H_k P_{k|k-1} H_k^T R_k^{-1}) z_k - K_k H_k \hat{x}_{k|k-1} \\ &= \hat{x}_{k|k-1} + (K_k + K_k H_k P_{k|k-1} H_k^T R_k^{-1} - K_k H_k P_{k|k-1} H_k^T R_k^{-1}) z_k - K_k H_k \hat{x}_{k|k-1} \\ &= \hat{x}_{k|k-1} + K_k (z_k - H_k \hat{x}_{k|k-1}) \end{aligned}$$

Summary

In summary the predictor equations are:

$$\begin{cases} \hat{x}_{k|k-1} &= \Phi_{k-1} \hat{x}_{k|k-1} && \text{mean} \\ P_{k|k-1} &= \Phi_{k-1} P_{k-1|k-1} \Phi_{k-1}^T + Q_{k-1} && \text{covariance} \end{cases}$$

² $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$

更新后的 PDF 的定义为 $P_{k|k}^{-1} \triangleq P_{k|k-1}^{-1} + H_k^T R_k^{-1} H_k$

$$P(x_k|Z_k) = \frac{\mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \mathcal{G}(x_k, \hat{x}_{k|k}, P_{k|k})}{\underbrace{\mathcal{G}(z_k, H_k \hat{x}_{k|k-1}, R_k + H_k P_{k|k-1} H_k^T) \int \mathcal{G}(x_k, \hat{x}_{k|k-1}, (P_{k|k-1} + H_k^T R_k^{-1} H_k)^{-1}) dx_k}_1}$$

$$P(x_k|Z_k) = \mathcal{G}(x_k, \hat{x}_{k|k}, P_{k|k})$$

最后我们必须找到更新后的估计值 $\hat{x}_{k|k}$ 。根据定义我们有

$$P_{k|k}^{-1} \hat{x}_{k|k} = P_{k|k-1}^{-1} \hat{x}_{k|k-1} + H_k^T R_k^{-1} z_k$$

为了获得 $\hat{x}_{k|k}$ ，我们使用矩阵求逆引理²找到 $P_{k|k}$

$$\begin{aligned} P_{k|k} &= \left[P_{k|k}^{-1} \right]^{-1} = (P_{k|k-1}^{-1} + H_k^T R_k^{-1} H_k)^{-1} \\ &= P_{k|k-1} - P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} H_k P_{k|k-1} \\ &= (I - P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} H_k) P_{k|k-1} \\ &= (I - K_k H_k) P_{k|k-1}, \quad K_k \triangleq P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} \end{aligned}$$

然后将 $P_{k|k}$ 和 $P_{k|k}^{-1} \hat{x}_{k|k}$ 相乘，我们得到

$$\begin{aligned} P_{k|k} P_{k|k}^{-1} \hat{x}_{k|k} &= (I - K_k H_k) P_{k|k-1} \left((P_{k|k-1}^{-1} \hat{x}_{k|k-1} + H_k^T R_k^{-1} z_k) \right) \\ \hat{x}_{k|k} &= (I - K_k H_k) \hat{x}_{k|k-1} + (I - K_k H_k) P_{k|k-1} H_k^T R_k^{-1} z_k \\ &= \hat{x}_{k|k-1} - K_k H_k \hat{x}_{k|k-1} + P_{k|k-1} H_k^T R_k^{-1} z_k - K_k H_k P_{k|k-1} H_k^T R_k^{-1} z_k \\ &= \hat{x}_{k|k-1} + (P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} (R_k + H_k P_{k|k-1} H_k^T) R_k^{-1} \\ &\quad - K_k H_k P_{k|k-1} H_k^T R_k^{-1}) z_k - K_k H_k \hat{x}_{k|k-1} \\ &= \hat{x}_{k|k-1} + (K_k (I + H_k P_{k|k-1} H_k^T R_k^{-1}) - K_k H_k P_{k|k-1} H_k^T R_k^{-1}) z_k - K_k H_k \hat{x}_{k|k-1} \\ &= \hat{x}_{k|k-1} + (K_k + K_k H_k P_{k|k-1} H_k^T R_k^{-1} - K_k H_k P_{k|k-1} H_k^T R_k^{-1}) z_k - K_k H_k \hat{x}_{k|k-1} \\ &= \hat{x}_{k|k-1} + K_k (z_k - H_k \hat{x}_{k|k-1}) \end{aligned}$$

Summary

总之，预测方程为：

$$\begin{cases} \hat{x}_{k|k-1} &= \Phi_{k-1} \hat{x}_{k|k-1} & \text{mean} \\ P_{k|k-1} &= \Phi_{k-1} P_{k-1|k-1} \Phi_{k-1}^T + Q_{k-1} & \text{covariance} \end{cases}$$

² $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$

And the update equations

$$\begin{cases} \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k(z_k - H_k\hat{x}_{k|k-1}) \\ P_{k|k} &= (I - K_k H_k)P_{k|k-1} \\ K_k &= P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} \quad \text{Kalman gain} \end{cases}$$

In order to obtain the results of section 6.3 we put the update equations in the predictor to find the prediction of the next state³

$$\begin{cases} \hat{x}_{k+1|k} &= \Phi_k \hat{x}_{k|k-1} + \Phi_k P_{k|k-1} H_k [H_k P_{k|k-1} H_k' + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1}) \\ P_{k+1|k} &= \Phi_k P_k \{I - H_k' [H_k P_{k|k-1} H_k' + R_k]^{-1} H_k P_{k|k-1}\} \Phi_k' + Q_k \end{cases}$$

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³In section 6.3 the notation P_{k+1} represents $P_{k+1|k}$ and is used for simplicity.

以及更新方程

$$\begin{cases} \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k(z_k - H_k\hat{x}_{k|k-1}) \\ P_{k|k} &= (I - K_k H_k)P_{k|k-1} \\ K_k &= P_{k|k-1} H_k^T (R_k + H_k P_{k|k-1} H_k^T)^{-1} \quad \text{Kalman gain} \end{cases}$$

为了获得 6.3 节的结果，我们将更新方程放入预测器中以找到下一个状态³的预测

$$\begin{cases} \hat{x}_{k+1|k} &= \Phi_k \hat{x}_{k|k-1} + \Phi_k P_{k|k-1} H_k [H_k P_{k|k-1} H_k' + R_k]^{-1} (z_k - H_k \hat{x}_{k|k-1}) \\ P_{k+1|k} &= \Phi_k P_k \{I - H_k' [H_k P_{k|k-1} H_k' + R_k]^{-1} H_k P_{k|k-1}\} \Phi_k' + Q_k \end{cases}$$

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³在第 6.3 节中，符号 P_{k+1} 代表 $P_{k+1|k}$ ，用于简单起见。

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